SPECIALIZED SCHEDULER AND NOTIFICATION SYSTEM

By

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ABSTRACT

Balancing physiotherapy sessions with academic responsibilities poses a significant challenge for students, particularly those living with physical disabilities, such as those attending King George VI center. To mitigate this challenge, we propose a Specialized Scheduler and Notification Mobile Application. This application enables seamless scheduling of therapy sessions for students and physiotherapists, ensuring alignment with academic schedules. By allowing students to input their academic timetable and physiotherapists to view students' available time slots, the system facilitates the scheduling of sessions during free periods. Automated reminders for upcoming sessions are sent to both students and physiotherapists, while students can track their session attendance progress. Developed using Java in Android Studio, this application offers personalized notifications, scheduling functionalities, and collaborative tools. It enhances time management, organization, and planning, while also supporting progress tracking, motivation, and communication. Ultimately, the proposed system aims to alleviate the challenges faced by students in balancing physiotherapy and academic commitments, fostering improved well-being and academic success.

Keywords: Scheduling, notifications, sessions, progress tracking, communication, appointment, physiotherapist

PREFACE

As we stand witness to the ever-evolving landscape of healthcare and education, it becomes increasingly apparent that certain segments of our community face unique challenges in accessing essential services. Among these challenges is the delicate balance between academic pursuits and healthcare needs, particularly for individuals living with physical disabilities. Within the halls of institutions like the esteemed King George VI center, where education and rehabilitation converge, the demand for innovative solutions to harmonize academic endeavors with physiotherapy regimens is paramount.

This Specialized Scheduler and Notification system emerged from a collective dedication to address the intricate needs of students navigating the intersection of academia and physical well-being. Grounded in the ethos of accessibility, efficiency, and empowerment, this system represents a collaborative effort to bridge the gap between education and healthcare, enhancing the overall quality of life for students with physical disabilities.

In the following pages, we embark on a journey through the intricacies of this system, exploring its functionalities, design principles, and overarching objectives. From its inception to its realization, this system embodies a commitment to excellence, driven by the imperative to facilitate seamless coordination between students and physiotherapists. It is our fervent hope that this system will serve as a beacon of support and resilience, empowering students to pursue their academic ambitions while nurturing their physical well-being.

We extend our deepest gratitude to all those who have contributed to the development of this system. May this endeavor stand as a testament to the transformative power of innovation and collaboration in fostering a more inclusive and equitable society.

ACKNOWLEDGEMENT

First and foremost, I would like to thank the Lord Almighty, whose presence and grace has been with me from the start of this program till now that l am achieving a great milestone.

I would like to thank my supervisor, Ms L Amos for her invaluable guidance, expertise, and continuous support throughout the entire research process. Her insightful feedback, constructive criticism, and encouragement have been instrumental in shaping and refining this project.

I am also immensely grateful to the faculty members of the Software Engineering department, whose teachings and mentorship have provided me with a strong foundation in my field of study. Their passion for academia and dedication to imparting knowledge have been truly inspiring.

I extend my heartfelt thanks to my mother and father for being prayer warriors and providing for me throughout the journey. I am also grateful to my family and friends for their unwavering support and belief in me. Their encouragement, patience, and understanding have been my pillars of strength during the ups and downs of this academic pursuit. I am grateful for their love, encouragement, and motivation that have kept me going.

I would like to acknowledge the participants of my study, whose willingness to share their experiences and insights has contributed significantly to the research findings. Their contributions have been invaluable in shaping the outcomes of this dissertation.

Lastly, I would like to express my profound appreciation to all the individuals who have indirectly contributed to this research through their academic work, publications, and previous studies. Their contributions have been instrumental in shaping the theoretical framework and methodology of this dissertation.

In conclusion, I am deeply thankful to everyone who has played a role, big or small, in the completion of this project. Your support, guidance, and encouragement have been invaluable, and I am truly grateful for the opportunity to have undertaken this research journey with your unwavering support.

DEDICATION

Dedicated to my dear sister Tinovimbanashe Marange and all physically challenged students at KG6,

Your resilience, strength, and unwavering determination have been a constant source of inspiration throughout the development of this system. Tinovimbanashe Marange, your courage in facing life's challenges with grace and fortitude serves as a guiding light, reminding us of the importance of empathy, accessibility, and inclusivity.

To all physically challenged students, this system is dedicated to you, your courage in overcoming obstacles, to your unwavering spirit in pursuing your dreams, and to your right to access essential services with dignity and ease. May this system serve as a tool to empower you, to support you, and to advocate for your needs in academic and healthcare settings.

This is to certify that HIT 400 Project entitled **Specialized Scheduler and Notification System has** been completed by **Tadiwanashe S Marange** (H200558M) for partial fulfilment of the requirements for the award of **Bachelor of Technology** degree in **Software Engineering**. This work is carried out by **her** under my supervision and has not been submitted earlier for the award of any other degree or diploma in any university to the best of my knowledge.

Ms L Amos	**Approved/Not Approved**
Ms L Amos	Project Coordinator
Signature: ……	**Signature: ……**
Date: ……	**Date: ……**

Certificate of Declaration

This is to certify that work entitled "**Specialized Scheduler and Notification System** "is *submitted in partial fulfillment of the requirements for the award of Bachelor of Technology (Hons) in Software Engineering, Harare Institute of Technology. It is further certified that no part of research has been submitted to any university for the award of any other degree.*

Ms L Amos	Signature……..	Date…….
(Mentor)	Signature……..	Date……
Mr Makondo	Signature……..	Date……..

Project Documentation Marking Guide

ITEM	TOTAL MARK /%	ACQUIRED/%
PRESENTATION- Format-Times Roman 12 for ordinary text, Main headings Times Roman 14, spacing 1.5. Chapters and sub-chapters, tables and diagrams should be numbered. Document should be in report form. Range of document pages. Between 50 and 100.Work should be clear and neat	**5**	
Pre-Chapter Section Abstract, Preface, Acknowledgements, Dedication & Declaration	**5**	
Chapter One-Introduction Background, Problem Statement, Objectives – smart, clearly measurable from your system. Always start with a TO… Hypothesis, Justification, Proposed Tools Feasibility study: Technical, Economic & Operational Project plan –Time plan, Gantt chart	**10**	
Chapter Two-Literature Review Introduction, Related work & Conclusion	**10**	
Chapter Three –Analysis Information Gathering Tools, Description of system Data analysis –Using UML context diagrams, DFD of existing system Evaluation of Alternatives Systems, Functional Analysis of Proposed System-Functional and Non-functional Requirements, User Case Diagrams	**15**	
Chapter Four –Design Systems Diagrams –Using UML Context diagrams, DFD, Activity diagrams Architectural Design-hardware, networking Database Design –ER diagrams, Normalized Databases Program Design-Class diagrams, Sequence diagrams, Package diagrams, Pseudo code Interface Design-Screenshots of user interface	**20**	
Chapter Five-Implementation & Testing Pseudo code of major modules /Sample of real code can be written here Software Testing-Unit, Module, Integration, System, Database & Acceptance	**20**	
Chapter Six –Conclusions and Recommendations Results and summary, Recommendations & Future Works	**10**	
Bibliography –Proper numbering should be used Appendices –templates of data collection tools, user manual of the working system, sample code, research papers	**5**	
	100	**/100**

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CHAPTER ONE: INTRODUCTION

1.0 Background

In educational institutions, particularly those accommodating students with physical challenges especially at King George VI center in Bulawayo, managing academic commitments alongside necessary medical treatments such as physiotherapy can be a complex task. Traditional scheduling systems often fall short in addressing the unique needs of these students, leading to conflicts between academic and therapeutic appointments, communication barriers, and inefficiencies in treatment adherence. Physiotherapy is critical for students with physical challenges, as it helps improve their mobility, strength, and overall quality of life. However, coordinating physiotherapy sessions around academic schedules can be challenging for both students and physiotherapists. Students may miss important classes or therapy sessions due to scheduling conflicts, while physiotherapists struggle to accommodate the diverse timetables of their patients. Moreover, existing systems lack effective communication tools, making it difficult for students to seek clarifications or updates from their therapists. Reminders for upcoming sessions are often manual and prone to oversight, leading to missed appointments

Recognizing these challenges, the development of a Specialized Scheduler and Notification System aims to bridge the gap between academic and therapeutic schedules, enhance communication, and ensure secure and efficient management of physiotherapy sessions. This system is designed to provide a holistic solution that not only simplifies scheduling but also supports the overall well-being and academic success of students requiring physiotherapy.

1.1 Problem statement

Students with physical challenges particularly at King George VI center often face significant difficulties in balancing their academic responsibilities with essential physiotherapy sessions. Traditional scheduling systems do not adequately account for the unique needs of these students, leading to several problems where existing scheduling systems fail to integrate academic and physiotherapy timetables, resulting in frequent conflicts. Students are often forced to choose between attending classes and their therapy sessions, compromising their academic performance and health outcomes. There is a lack of effective communication tools within current systems, making it difficult for students to interact with their physiotherapists. Missed Appointments, without automated reminders, students and physiotherapists often forget scheduled sessions.

Addressing these issues, the Specialized Scheduler and Notification System aims to provide a comprehensive solution that integrates academic and physiotherapy schedules, enhances communication, ensures timely reminders and facilitates progress tracking. This system is essential to improve the overall management of physiotherapy sessions, supporting the academic and health needs of students with physical challenges.

1.2 Aim

The aim of this project is to develop a Specialized Scheduler and Notification Application that specifically improves physiotherapy session adherence for students with disabilities compared to traditional methods, as measured by attendance data tracked within the application.

1.2.1 Objectives

1. To design a user-friendly interface for appointment scheduling that specifically considers the academic schedules of the student to avoid conflicts with physiotherapy sessions.
2. To allows the physiotherapists to schedule sessions with their patients/ students
3. To allows students to easily communicate with their physiotherapist for clarifications, questions, or updates
4. To allow automated reminders through email, or push notifications to ensure students and physiotherapist are reminded of their upcoming physiotherapy sessions.
5. To create exercise plans tailored to each student's needs which they can share with their patients
6. To allows physically challenged students to securely login and access their personalized physiotherapy exercise plans as shared by their physiotherapist
7. To allow users to track and monitor their physiotherapy attendance progress to enable adjustments to their physiotherapy attendance schedules.

1.3 Hypothesis

The implementation of a specialized scheduler and notification system tailored to the needs of students living with physical disabilities at King George VI center will lead to improved accessibility and effectiveness of physiotherapy services, resulting in:

- Increased adherence to physiotherapy sessions: The system will provide personalized scheduling capabilities, reminders, and notifications, facilitating better adherence to scheduled physiotherapy sessions among students.
- Improved health outcomes: With better adherence to physiotherapy sessions and enhanced coordination, students will experience improved physiotherapy outcomes, including increased mobility, reduced pain, and enhanced overall well-being.
- Enhanced user satisfaction: The system's user-friendly interface, personalized features, will contribute to higher levels of user satisfaction among students and healthcare providers.
- Reduction in missed appointments: Through automated reminders and notifications, the system will significantly reduce the occurrence of missed appointments.
- Positive impact on academic performance: Improved health and well-being resulting from better access to physiotherapy services will positively impact students' academic performance, leading to higher levels of engagement, productivity, and success in their academic pursuits.

1.3.1 Justification

A Scheduler and Notification Application for students living with physical disabilities at King George VI is justified as it addresses their unique needs, supports time management, ensures accessibility and inclusivity, improves organization and planning, enables progress tracking and motivation, and enhances communication and support

- The system enhances accessibility to physiotherapy services by streamlining the scheduling process and providing intuitive interfaces that cater to diverse needs. Through features such as mobile compatibility, assistive technologies, and customizable preferences, the system ensures that students with disabilities can easily access and manage their physiotherapy appointments.
- Students living with disabilities may require additional time and effort to complete their school work and attend physiotherapy sessions. The system helps them allocate time slots for different activities, set reminders, and avoid scheduling

conflicts. This support enables better time management, reducing stress and improving productivity.

- Automated reminders and notifications provided by the system help students stay on track with their physiotherapy sessions, reducing the likelihood of missed appointments and improving adherence to treatment plans. This proactive approach to scheduling and reminders fosters a culture of accountability and commitment to healthcare among students.
- By automating manual processes and reducing administrative burdens, the system improves the efficiency of physiotherapy services at King George VI center. Healthcare providers can streamline appointment scheduling, optimize resource allocation, and focus more on delivering quality care to students.
- A dedicated messaging platform within the application facilitates seamless communication between students, physiotherapists. Students can seek clarifications, ask questions, and receive updates, ensuring they have the necessary support to balance their school work and physiotherapy effectively. This enhanced communication fosters collaboration and a supportive environment.

1.3.2 Proposed tools

1. Programming language - Java
2. Integrated Development Environment (IDE) – Android Studio
3. Database – MySQL
4. UX/ UI design – figma
5. API – google calendar API
6. Testing and Debugging – Android Virtual Device

1.4 Feasibility study

The Specialized Scheduler and Notification application is designed to help students living with physical disabilities balance between school work and physiotherapy sessions. This feasibility study aims to determine the viability of developing such an application.

1.4.1 Technical Feasibility

The development of the Specialized Notification and Scheduler Application requires the use of modern software development tools and technologies, including mobile application development

frameworks, calendar integration APIs, push notification services, database management systems, and collaboration and communication tools. All of these tools are readily available and widely used in the industry, making the project technically feasible.

1.4.2 Economic Feasibility

The development of the Specialized Notification and Scheduler Application will require a significant investment of time and resources. The project requires a team of developers, designers, and project managers, as well as hardware and software infrastructure, these are already available as the researcher is going to develop the project and the hardware and software requirements are open source. A cost benefit analysis indicates that the initial investments for the project is justified by the potential benefits of the application, including improved academic performance, improved attendance of physiotherapy sessions ensuring well-being of physically challenged students living with disabilities. The projected return on investment (ROI) is estimated to exceed the project costs within the first two years of implementation.

1.4.3 Operational Feasibility

The Specialized Notification and Scheduler Application is designed to be user-friendly and intuitive, with features that are easy to use and understand. The application is easily accessible and inclusive ensuring that all students, regardless of their disabilities, can use the application effectively. The application integrates seamlessly with existing calendar applications, making it easy for students to adopt and use. Therefore, the project is operationally feasible.

1.4.4 Legal and Ethical Feasibility

The development of the Specialized Scheduler and Notification application must comply with legal and ethical requirements. The Specialized Scheduler and Notification adheres to data protection laws, ensuring that student data is kept secure and confidential. The application is designed to respect the privacy of students. Additionally, the application is designed to be inclusive and non-discriminatory, ensuring that it does not violate any ethical standards. Therefore, the project is legally and ethically feasible.

1.5 Project plan

Figure 1 Time Plan

Task	Start	Finish	Duration (Weeks)
Project Proposal	2/09/2023	08/09/2023	1week

Requirements Gathering	10/09/2023	30/09/2023	3 weeks
System Analysis	1/10/2023	14/10/2023	2weeks
System Design	17/10/2023	13/11/2023	4 weeks
System Development	15/11/2023	28/04/2024	24 weeks
System Testing	24/11/2024	30/04/2024	26 weeks
System Implementation	17/11/2023	01/05/2024	2weeks
Documentation	2/09/2023	28/05/2024	36 weeks

Figure 2 Gantt Chart

Software Development lifecycle activities	September	October	November	December	January	February	March	April	May
Project proposal and planning									
Requirements gathering and analysis									
System Design									
System Development									
System testing and validation									
Implementation									
Maintenance									

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The aim of this literature review is to explore the existing research and scholarly work related to the development and implementation of a Specialized scheduler and notification application for assisting disabled students in balancing their school workload and managing their physiotherapy sessions. This review seeks to provide a comprehensive understanding of the current state of knowledge, identify gaps in the existing literature, and gain insights into the challenges and opportunities associated with such an application.

2. 2 Related Work

The SnoopMe application[1] is designed to assist students in managing their tasks effectively by providing alerts and monitoring their mobile usage. It allows students to schedule tasks, set reminders, and receive notifications about task completion. The app tracks mobile activities and recommends usage restrictions based on the data collected. Students can benefit from graphical representations of their time spent on different apps. Feedback from users indicates positive experiences with the app, with suggestions for additional features to enhance its effectiveness as a student-centric scheduler.

Meeting Scheduler using unpleasant notification [2], the system employs a progressive strategy involving unpleasant notifications to prompt users to promptly input their schedules. Initially, users receive a standard notification, like an email, requesting schedule input. In cases of non-compliance, the system intensifies the notifications, such as informing the user's superior about the delay or publicly listing non-compliant users. This gradual escalation of unpleasantness serves to motivate users to swiftly input their schedules to avert additional negative repercussions. The system has the following benefits i) encouraging prompt input of schedules, by employing unpleasant notifications, users are motivated to input their schedules quickly to avoid unpleasant consequences. ii)Improved meeting scheduling efficiency iii) the system helps eliminate delays caused by participants not inputting their schedules promptly. iv)Increased accountability, users are held accountable for inputting their schedules, leading to more reliable and efficient meeting arrangements v) Enhanced user engagement, the use of various notification methods can increase user awareness and engagement in the scheduling process.

In [3] the thesis focuses on designing a Meeting Scheduling System (MSS) to facilitate inter-organizational meeting scheduling. It addresses the challenges of coordinating meetings across different organizations by automating the process of sharing calendar information and negotiating meeting times. The software architecture of MSS allows for effortless integration with heterogeneous calendar systems, aiming to streamline the scheduling process. The system emphasizes a balance between automation and user control to maximize satisfaction and efficiency. Future research suggestions include expanding integration support for various calendar systems and studying the integration of MSS with large external CRM systems.

In [4] an android application, focusing on enhancing interactions between doctors and patients while promoting adherence to medication schedules. The application includes a feature that automatically sets off alarms to remind users of their medication timings. Additionally, it allows patients to search for healthcare providers, access health-related information, and receive notifications via email or messages. The system aims to be cost-effective, efficient, and supportive of medication adherence by offering features like facilitating communication between doctors and patients, assisting in medication scheduling, and presenting a user-friendly interface that does not necessitate specialized hardware for operation.

The Project Tracker and Organizer software is a tool designed to assist in assigning project modules to employees and tracking project progress efficiently[5]. Developed by Krishna Kanth Kambhampaty for a Master of Science degree in Computer Science at North Dakota State University, this tool simplifies project scheduling and monitoring. It offers features such as data entry through interactive forms, data updating capabilities for administrators, reliable system performance with fast page loads, centralized database for quick data retrieval, and automatic report generation for project management. The software follows a modified waterfall model for development and can be accessed with minimal hardware requirements. Future work suggestions include graphical representations, scalability for larger projects, progress bars for project status, email notification system, and the integration of intelligent heuristic algorithms for project scheduling. developed using Visual Studio .Net 2012 as the development framework. The programming language used was C#, and ASP.NET was utilized as the server-side web application framework. The backend database chosen for this application was MS SQL Server 2010 (SQL Server 2008 R2).

The Mobile Application for Personal Activities Management (Life Scheduler)[6] aims to help users organize their tasks efficiently. Key features include automatic task organization, alerting on one task at a time to maintain focus, and the use of an arrangement algorithm to prioritize tasks sequentially. The app merges management and AI sciences to facilitate task organization, allowing users to concentrate on various aspects of their lives. Life Scheduler introduces a unique arrangement algorithm to streamline task execution. The Mobile Application for Personal Activities Management (Life Scheduler) integrates principles from management and artificial intelligence (AI) to streamline task organization. By leveraging these disciplines, the app assist users in efficiently managing their tasks, enabling them to focus on different facets of their lives without the burden of manual task organization. The project utilized Extreme Programming as a software development methodology due to the team's small size and the rapid requirements changing nature of the project.

In [7] is a mobile application for scheduling team meetings, notification of meetings, monitoring the team that is conducting the meeting was developed. The mobile application addresses issues related to monitoring, notifications, and multi-device access. The mobile application is also used for data collection by the company admin. It utilizes features like location tracking and push notifications. The application's user interface is designed to be responsive across various devices. The method used in making this application was the prototype method for gathering needs, designing and evaluating programs. For problem analysis method uses Fishbone analysis to obtain specific problems. The mobile application was tested using the black box method to ensure the system runs well.

The InvigilatorApp[8], a system designed for efficient scheduling and reminder notifications for invigilation duties in educational institutions. The system automate and optimize the scheduling process, improve communication between administrators and invigilators, and streamline exam administration. Key features include web-based admin and faculty dashboards, real-time notifications, round-robin scheduling, and database management. The system utilizes algorithms such as Particle Swarm Optimization, Constraint Programming, Ant Colony Optimization, Tabu Search, and Hybrid Approaches to enhance scheduling efficiency. The system architecture involves React JavaScript for the user interface and Django framework for the backend, ensuring timely reminders and email notifications.

The Smart Meeting Room Scheduling and Management System [9]utilizes real-time occupancy detection to optimize room utilization and support ad-hoc meetings. The system includes PIR sensor fusion devices mounted in meeting rooms, connected to embedded modules with Ethernet connectivity. These modules send occupancy status data to a local server via UDP protocols. A scheduling application on the server manages meeting room bookings and monitors real-time occupancy status. The system conducts sensor tests, network tests, and system tests to ensure accuracy and functionality. Users can access the scheduling application from smartphones, tablets, and PCs to book rooms and check real-time availability for ad-hoc meetings. The system automatically sends emails to meeting organizers if occupancy policies are violated.

2.3 Conclusion

The related works highlights the importance of user-friendly interfaces, automated reminders, and efficient communication channels in enhancing productivity and adherence to schedules. Systems like SnoopMe, Meeting Scheduler using unpleasant notification, and Life Scheduler underscore the significance of personalized alerts and notifications to motivate users and improve engagement. Moreover, the integration of advanced algorithms and technologies, as seen in the InvigilatorApp and Smart Meeting Room Scheduling and Management System, showcases the potential for optimizing scheduling processes and enhancing user experiences through real-time data analysis and system automation. From the reviewed literature it can be seen that there is a need for tailored solutions that cater specifically to the unique requirements and challenges faced by disabled/ physically challenged students in managing both academic commitments and physiotherapy sessions. Additionally, the reviewed literature emphasizes the importance of accessibility, security, and inclusivity in designing such applications to ensure equitable access and user satisfaction.

By leveraging insights from existing systems and addressing identified gaps, the proposed Specialized Scheduler and Notification Application has the potential to significantly improve the academic and healthcare experiences of disabled students, fostering productivity, and well-being. Further research and development efforts in this area are warranted to realize the full potential of specialized scheduling solutions for disabled individuals.

CHAPTER THREE: ANALYSIS

3.1 Introduction

This chapter focuses on requirements analysis. It also describes how the current system works, its faults and drawbacks. This chapter also includes the functional requirements and capabilities that the new proposed system will provide.

3.2 Information gathering tools

3.2.1 Interview

Interview with the physiotherapist

1. Can you provide an overview of your experience as a physiotherapist?
2. Can you describe your current process for scheduling physiotherapy sessions with patients/students?
3. Are there any specific pain points or inefficiencies in the current scheduling process that you would like to address?
4. How do you currently communicate with your patients/students for clarifications, questions, or updates?
5. Are there any communication barriers or challenges you face in your interactions with patients/students?
6. Do you use any software or tools to assist you in your practice as a physiotherapist?
7. What specific features or functionalities would you like to see in a system for scheduling physiotherapy sessions?
8. Automated Reminders:
9. How do you currently remind patients/students about upcoming physiotherapy sessions?
10. How do you currently create exercise plans for your patients/students?
11. How do you currently track and monitor the progress of your patients/students during physiotherapy treatment?
12. What information or metrics do you typically track to assess progress?

3.2.2 Questionnaire

Please provide answers by ticking in the box or filling in information in the provided space.

1. How important is it for the system to have a user-friendly interface for appointment scheduling?

   Very important ☐

   Important ☐

   Neutral ☐

   Not important ☐

2. What specific features would you expect in the user interface to facilitate appointment scheduling?

   ---

3. How often do you encounter conflicts between academic schedules and physiotherapy sessions?

   Frequently ☐

   Occasionally ☐

   Rarely ☐

   Never ☐

4. How do you currently schedule sessions with your physiotherapist?

   Manually through phone calls or emails ☐

   Using a scheduling software ☐

   Other (please specify) ----------

5. What features would you find most beneficial in a system for scheduling sessions with your physiotherapist?

   ---

6. How do you currently communicate with your physiotherapist for clarifications, questions, or updates?

   In-person appointments ☐

Phone calls ☐

Emails ☐

Other (please specify) ----------

7. How do you currently receive reminders for upcoming physiotherapy sessions?

Phone calls ☐

Emails ☐

Text messages ☐

Other (please specify) ----------

8. How do you currently track and monitor your physiotherapy progress?

Written logs ☐

Verbal updates with the physiotherapist ☐

Other (please specify) ----------

3.3 Description of system

In the existing system, scheduling between students and physiotherapists at King George VI center is primarily manual, with students visiting the physiotherapist at their own discretion and free time. There isn't a centralized digital platform or application facilitating appointments or coordination between the two parties. Instead, students independently arrange appointments with the physiotherapist based on their availability and needs, through in-person visits. Then they can use any digital calendar if they want where both students and physiotherapists schedule their activities on their own phones which they will be reminded of. This manual approach lacks automation and real-time scheduling updates, relying solely on direct communication between students and physiotherapists for appointment management.

3.4 Data Analysis

3.4.1 Context Diagram

Figure 3 Student Level 0 DFD

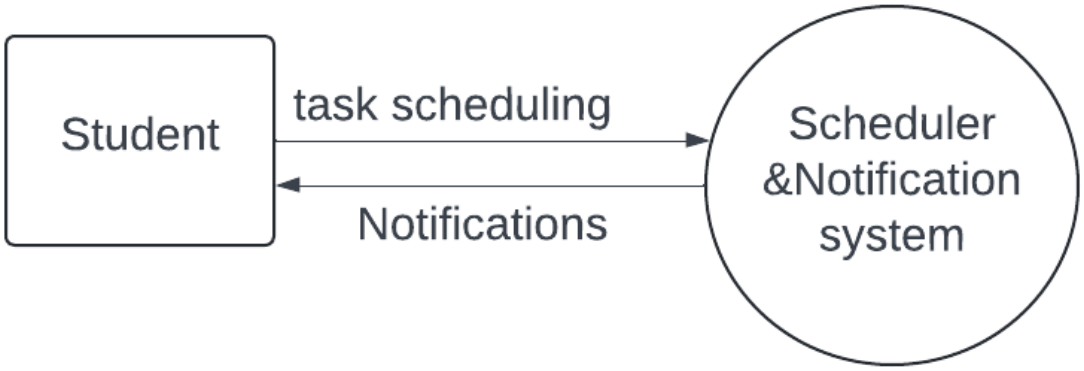
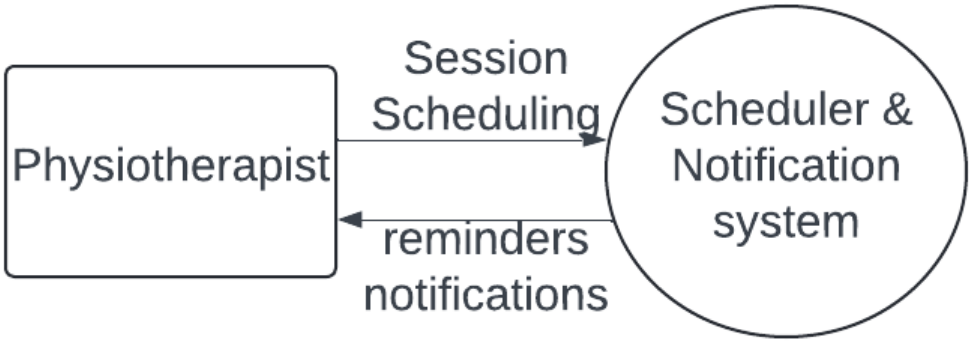


Figure 4 Physiotherapist Level0 DFD



3.4.2 Data Flow Diagram

Figure 5 P.C Student - Level 1 DFD

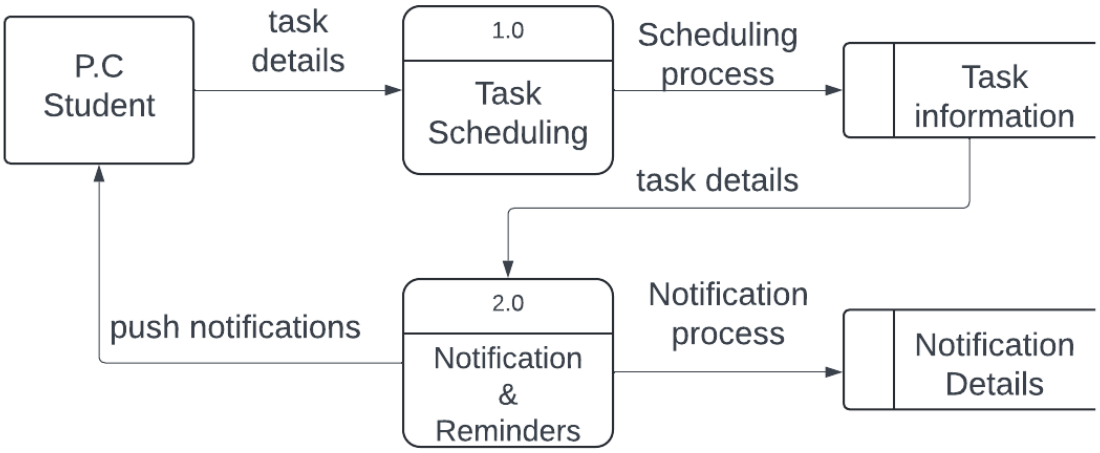
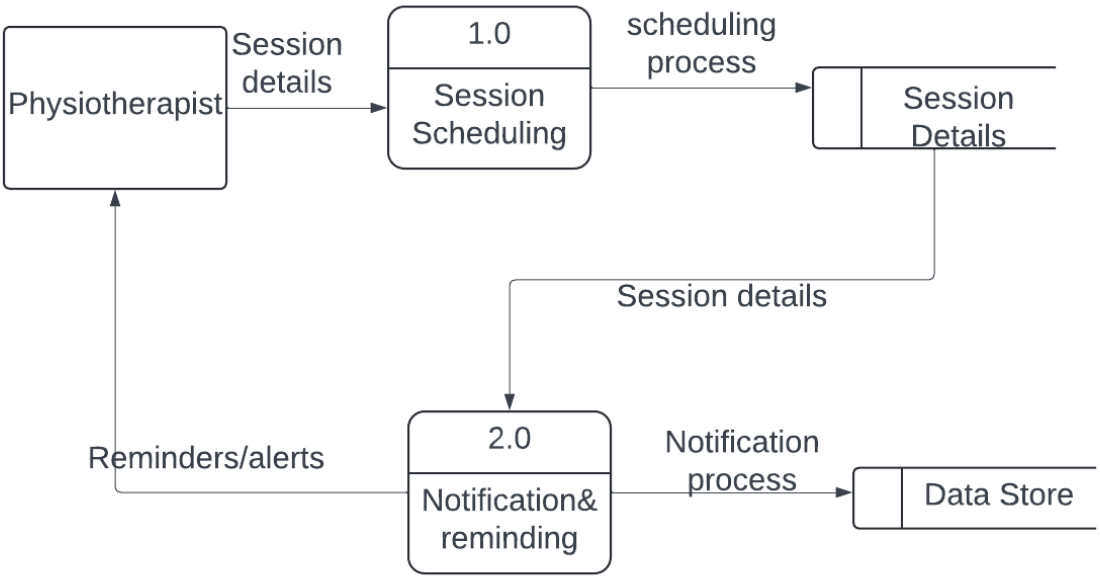


Figure 6 Level 1 DFD for Physiotherapist



3.5 Evaluation of alternatives systems

There are several alternative systems for a scheduler and notification application that could be useful for students living with physical disabilities to balance their school work and physiotherapy.

1. Google Calendar - Google Calendar is a widely used scheduling application that can be accessed from any device with internet access. It allows users to create and manage multiple calendars, set reminders and notifications, and share their schedules with others. Students could use this application to schedule their physiotherapy appointments and set reminders for school assignments and deadlines.

2. My Study Life - My Study Life is a free scheduling application designed specifically for students. It allows users to input their class schedules, assignments, and exams, and set reminders for each. The application also includes a feature to track progress and grades. Students could use this application to schedule their physiotherapy appointments around their class schedule and make sure they don't miss any important deadlines.

3. Trello - Trello is a project management and collaboration tool that can be used to create schedules and to-do lists. It allows users to create boards and cards for different tasks and set due dates and reminders for each. Students could use Trello to create a board for their school work and physiotherapy appointments, and set reminders to ensure they stay on track.

3.6 Functional Analysis of Proposed System

3.6.1 Functional Requirements

1. User Registration
    - The system shall allow users (students and physiotherapists) to register accounts.
    - Users shall be able to log in securely using credentials (e.g., username and password).
2. Academic Schedule Input
    - Students shall be able to input their academic schedules, including class times, breaks, and extracurricular activities.
3. Session Scheduling:
    - Physiotherapists shall be able to view students' available time slots based on their academic schedules.
    - Physiotherapists shall be able to schedule therapy sessions for students at their free periods, ensuring no conflicts with academic commitments.

4. Automated Reminders
    - The system shall send automated reminders to both students and physiotherapists prior to scheduled therapy sessions.
    - Reminders shall include session details (date, time)
5. Progress Tracking
    - Students shall be able to track their attendance and progress in therapy sessions.
    - The system shall provide visual representations of progress, such as charts or graphs, for easy interpretation.
6. Notification System
    - The system shall support personalized notifications for various events, including session reminders, schedule updates, and progress milestones.
    - Notifications shall be delivered via push notifications or email.
7. Communication and Collaboration
    - The system shall facilitate communication and collaboration between students and physiotherapists.
    - Users shall be able to send messages, share documents or notes, and request clarifications or updates regarding therapy sessions.

3.6.2 Non-Functional Requirements

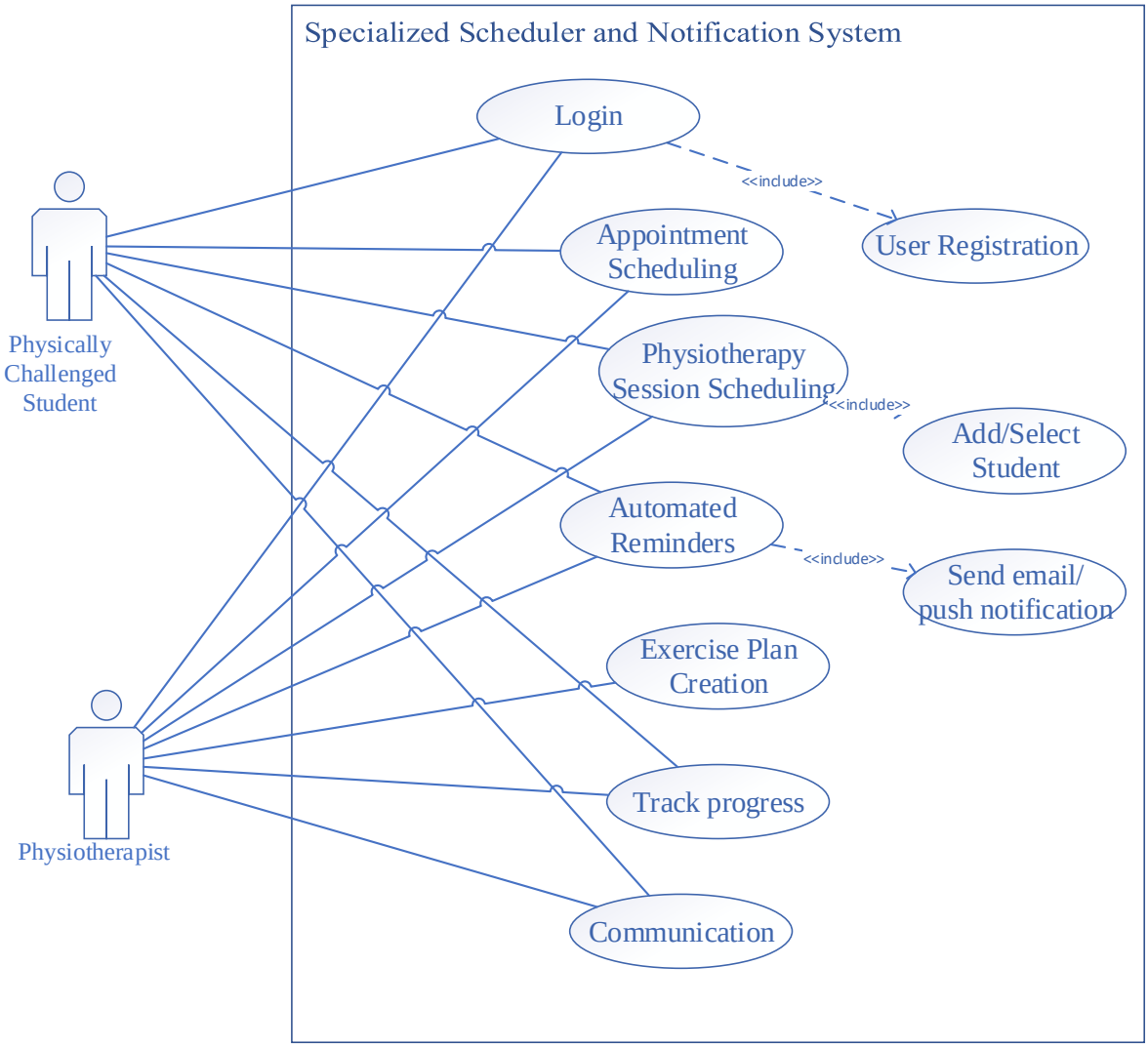
1. Performance
    - The system shall respond to user interactions within acceptable response times, even under peak loads.
    - Session scheduling and notification delivery shall occur promptly and without significant delays
2. Reliability
    - The system shall be available and accessible to users without frequent downtime.
    - Automated reminders and notifications shall be reliable and delivered consistently.
3. Usability
    - The user interface shall be intuitive and easy to navigate, catering to users of varying technological proficiency.

- Accessibility features shall be implemented to ensure usability for individuals with disabilities.

4. Compatibility
    - The system shall be compatible with a wide range of devices and operating systems, including smartphones, tablets, and desktop computers.
5. Scalability
    - The system shall be capable of accommodating an increasing number of users and data without degradation in performance.
    - It should scale seamlessly as the user base grows or as additional features are introduced
6. Security
    - User data shall be encrypted during transmission and storage to prevent unauthorized access.
7. Compliance
    - The system shall comply with relevant industry standards and regulations, including data protection laws and accessibility guidelines.
    - Regular audits and assessments may be conducted to ensure ongoing compliance.

3.7 Use Case Diagram

Figure 7 Use case diagram

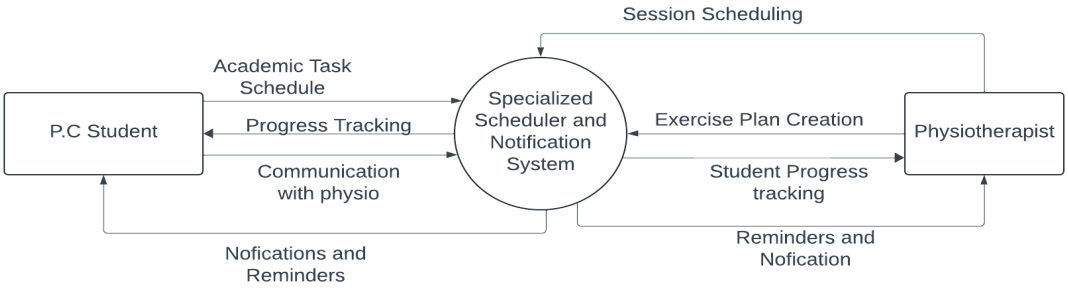


CHAPTER FOUR: DESIGN

4.1 System Diagrams

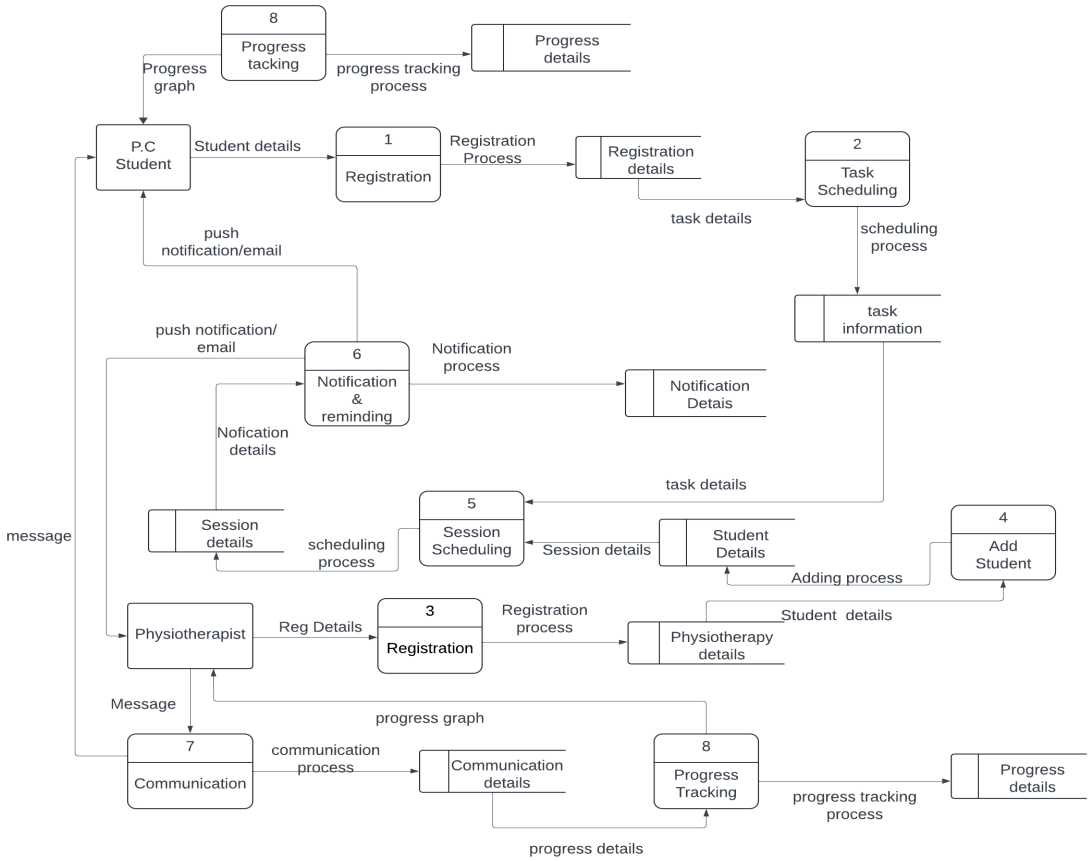
4.1.1 Context Level DFD

Figure 8 Level 0 DFD



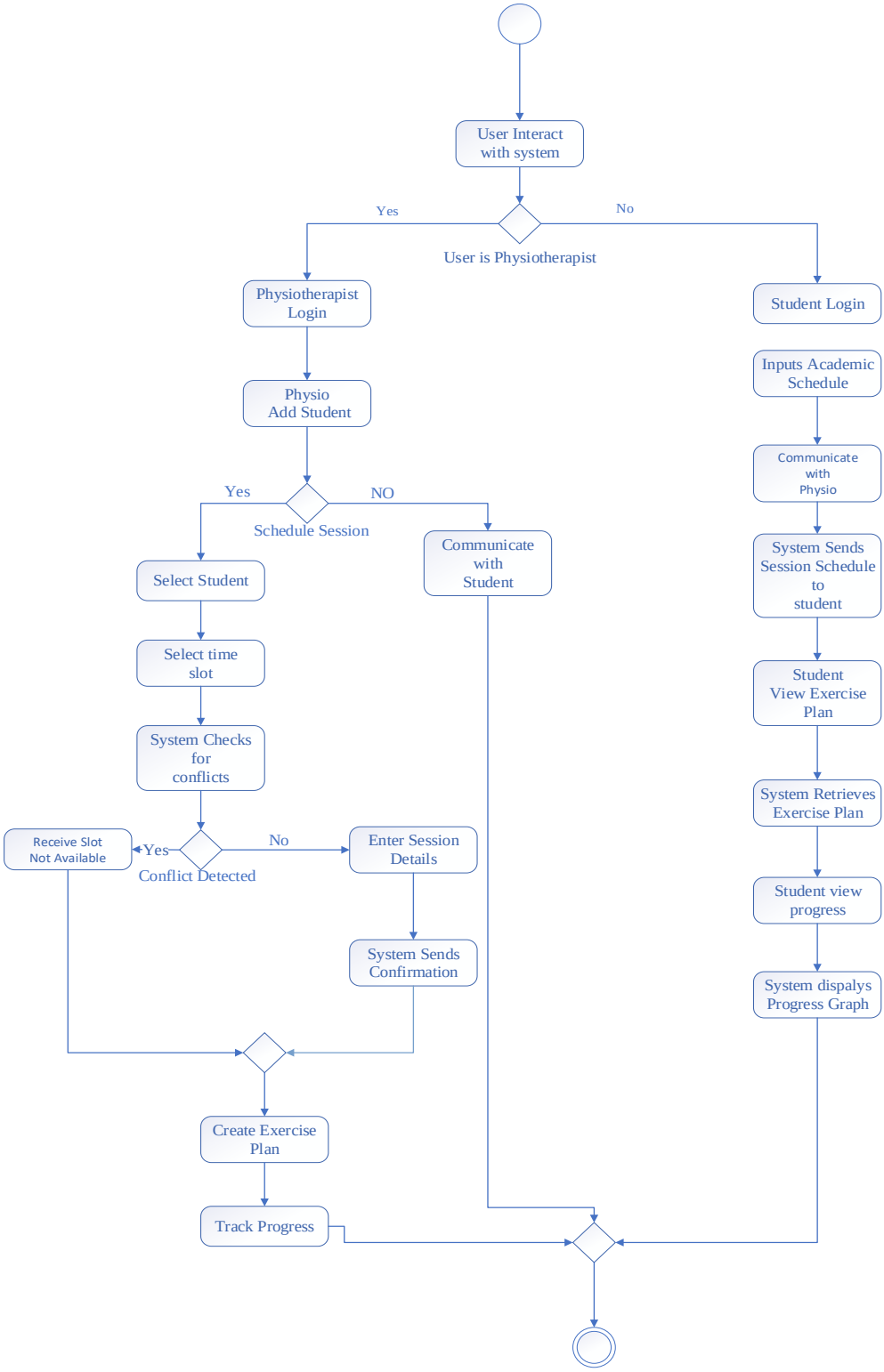
4.4.2 Data Flow Diagram

Figure 9 Level 1 DFD



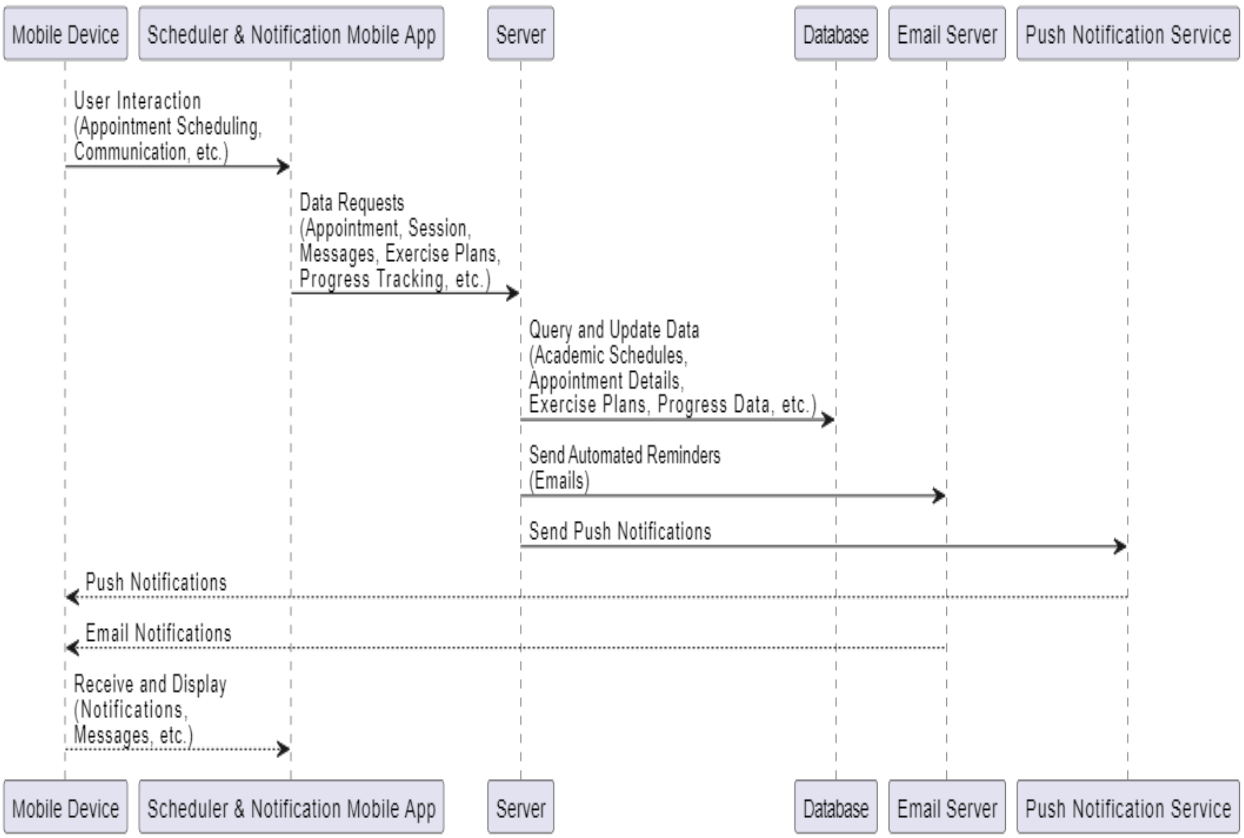
4.1.3 Activity Diagram

Figure 10 Activity Diagram



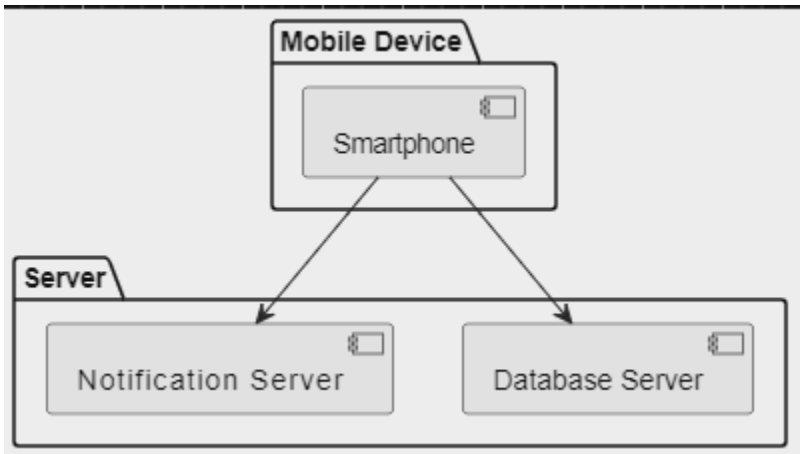
4.2 Architectural Design

Figure 11 Architectural Solution



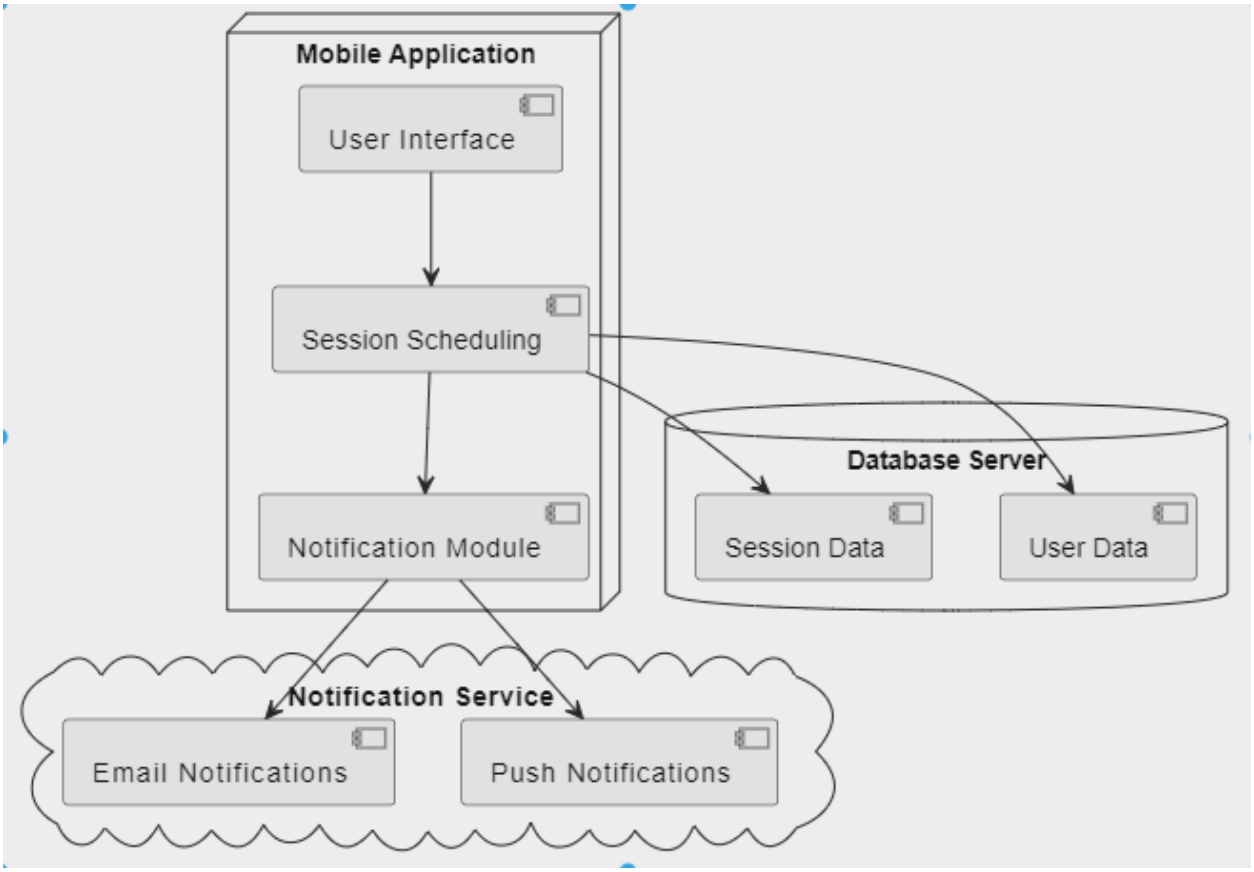
4.2.1 Hardware Design

Figure 12 Hardware Diagram



4.2.2 Network Design

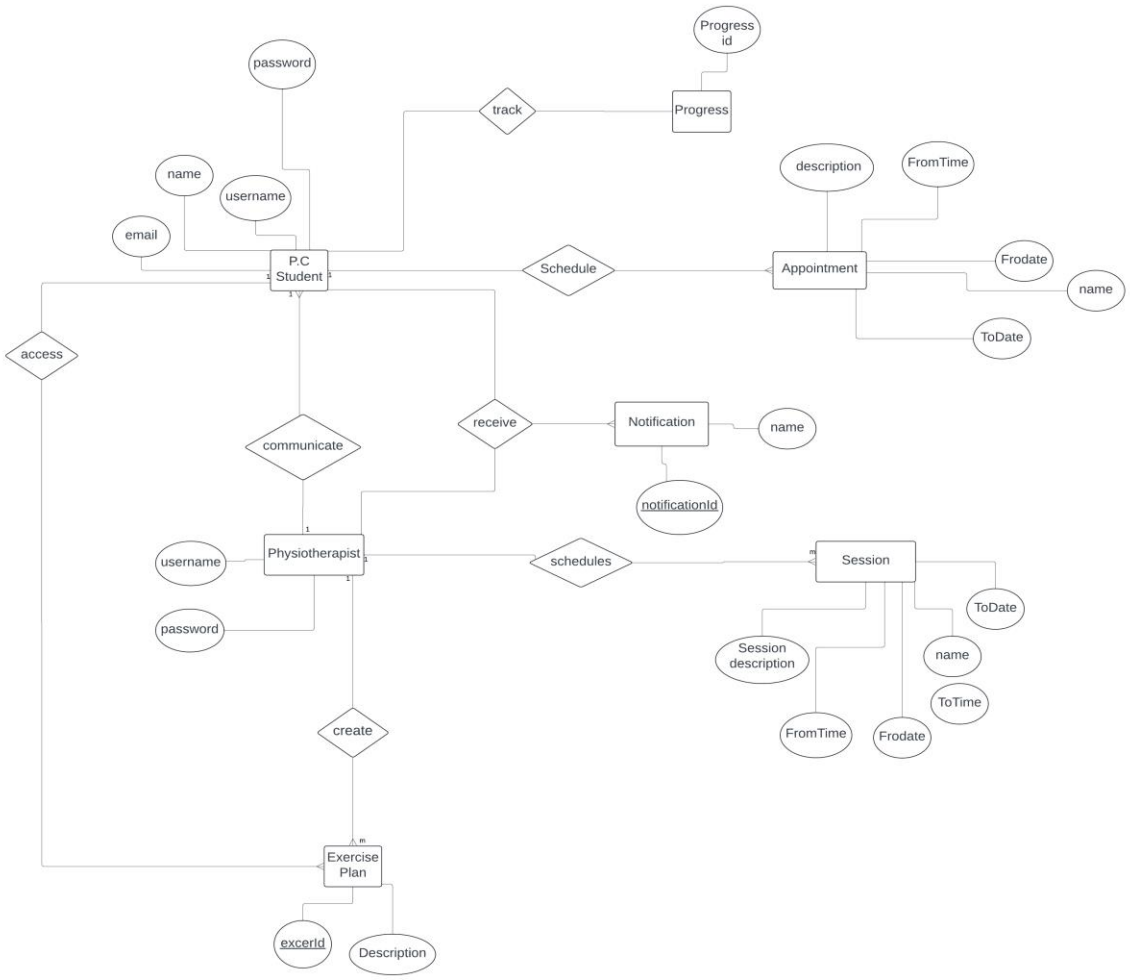
Figure 13 Networking Diagram



4.3 Database Design

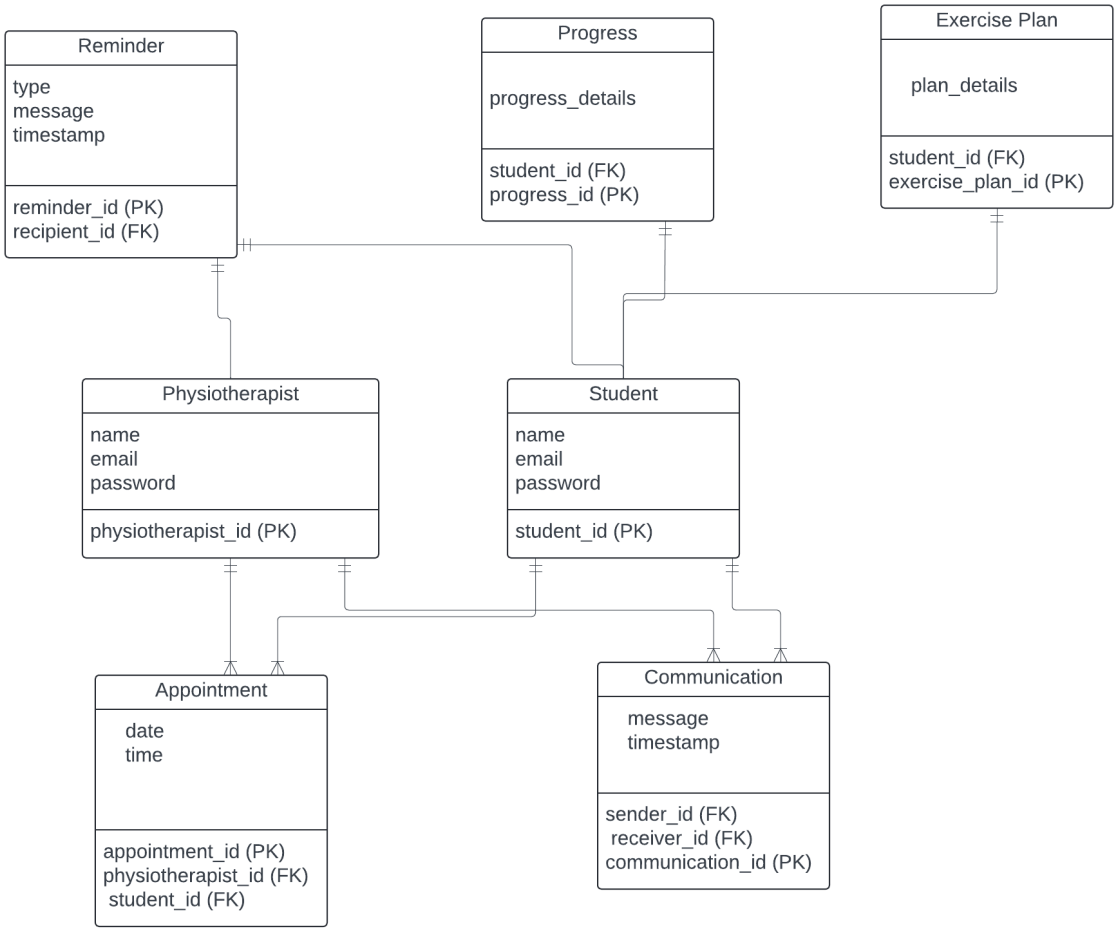
4.3.1 Entity Relation Diagram

Figure 14 ERD



4.3.2 Normalized Database

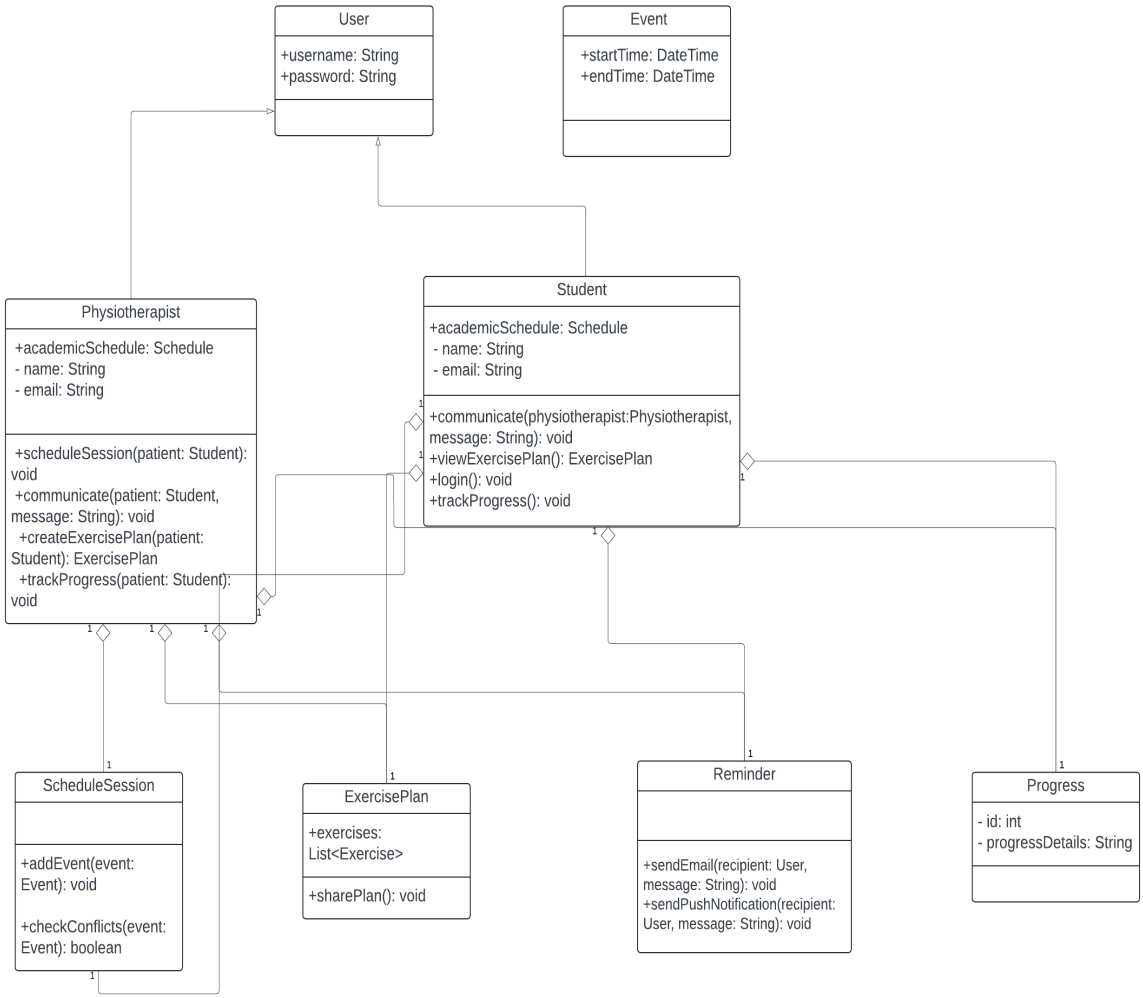
Figure 15 Normalized Database



4.4 Program Design

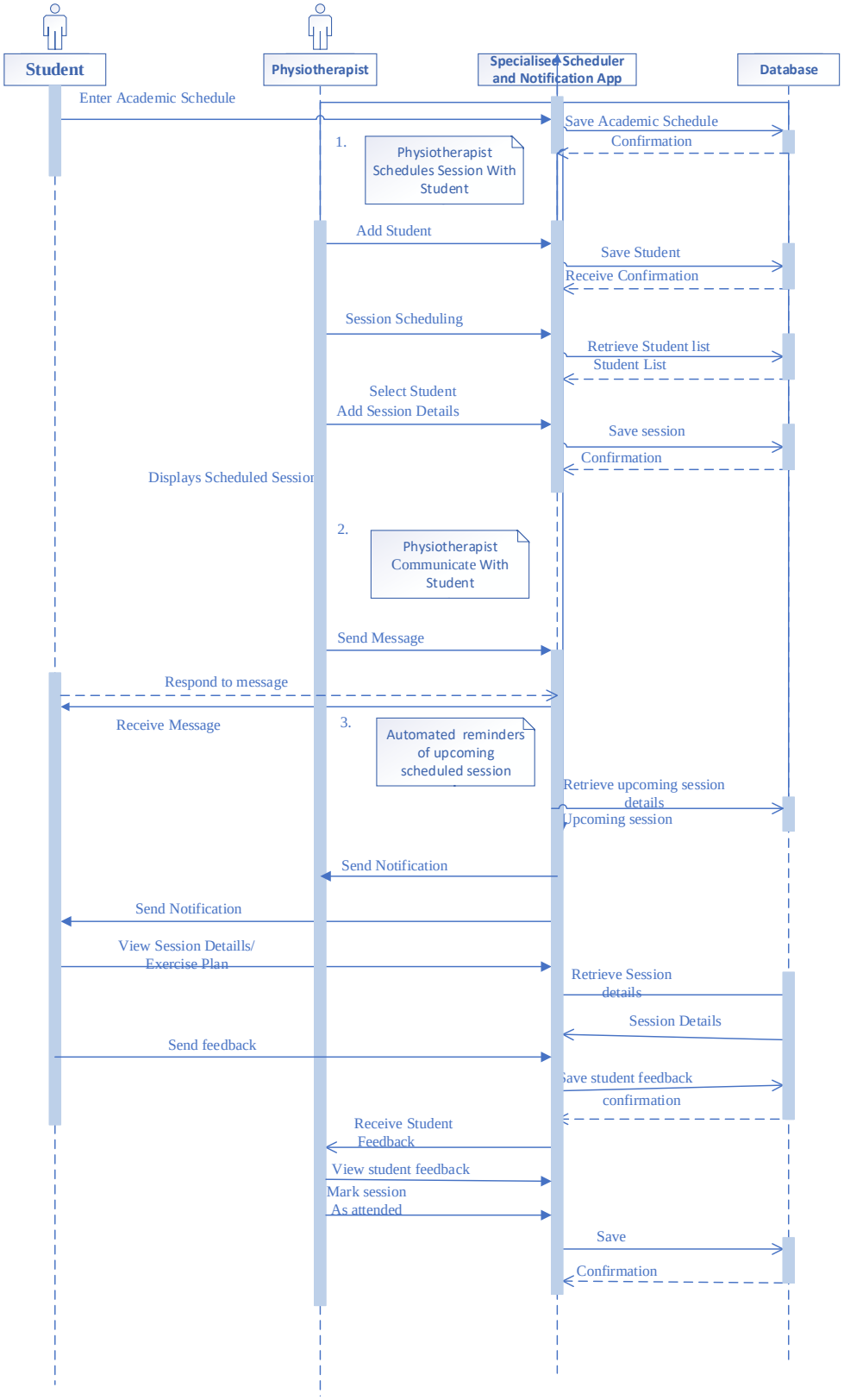
4.4.1Class diagram

Figure 16 Class Diagram



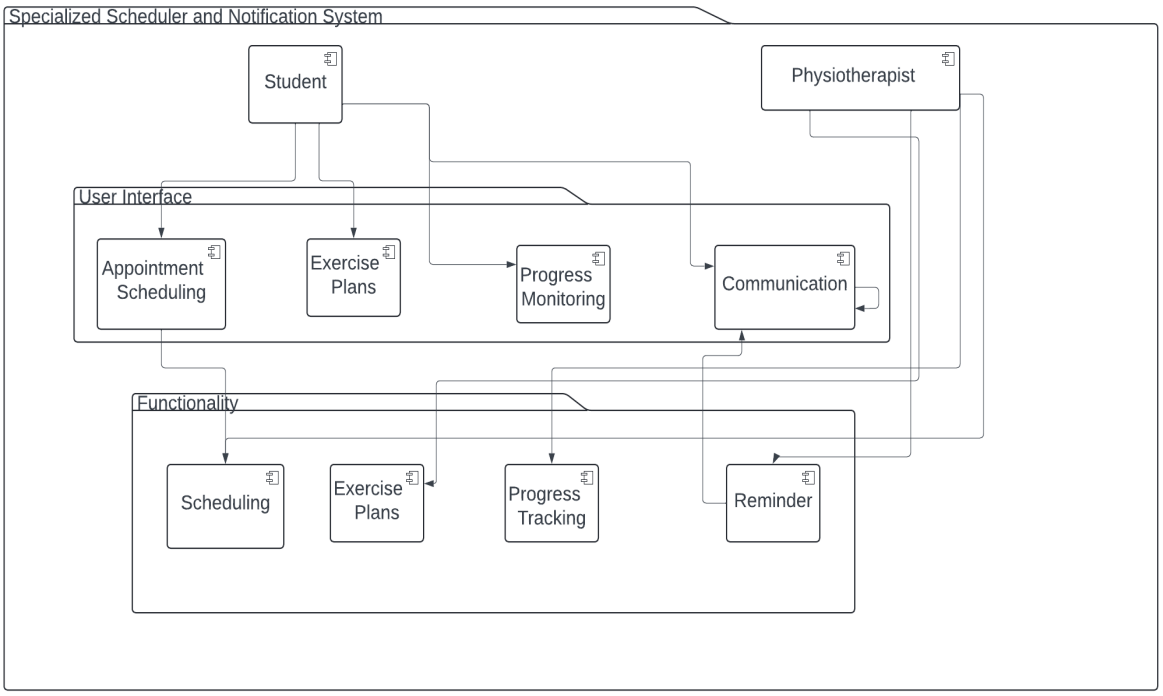
4.4.2 Sequence Diagram

Figure 17 Sequence Diagram



4.4.3 Package Diagram

Figure 18 Package Diagram



4.4.4 Pseudocode

<u>1. Pseudocode for Appointment Scheduling</u>

```
FUNCTION scheduleAppointment(student, physiotherapist, dateTime):
  IF isDateTimeValid(dateTime) THEN
    IF isAvailable(student, dateTime) THEN
      appointment = createAppointment(student, physiotherapist, dateTime)
      addToCalendar(appointment)
      RETURN SUCCESS
    ELSE
      RETURN "Student is not available at the specified time."
    END IF
  ELSE
```

```
        RETURN "Invalid date and time."
    END IF
END FUNCTION
```

2. Pseudocode for Automated Reminders

```
FUNCTION sendReminder(appointment):
    reminderTime = appointment.dateTime - reminderThreshold
    IF currentDateTime >= reminderTime THEN
        IF appointment.student.notificationPreference == "email" THEN
            sendEmail(appointment.student.email, "Reminder: Upcoming Appointment", "Dear [Student Name], You have an appointment scheduled with [Physiotherapist Name] on [Date and Time].")
        ELSE IF appointment.student.notificationPreference == "push notification" THEN
            sendPushNotification(appointment.student.deviceToken, "Reminder: Upcoming Appointment", "You have an appointment scheduled with [Physiotherapist Name] on [Date and Time].")
        END IF
    END IF
END FUNCTION
```

3.Pseudocode for Communication

```
FUNCTION sendMessage(sender, receiver, message):
    IF isValidUser(sender) AND isValidUser(receiver) THEN
        IF isUserAuthorized(sender) AND isUserAuthorized(receiver) THEN
            IF isMessageValid(message) THEN
                IF isConnectionEstablished(sender, receiver) THEN
                    storeMessage(sender, receiver, message)
                    RETURN SUCCESS
                ELSE
                    RETURN "Unable to establish communication with the receiver."
```

```
            END IF
        ELSE
            RETURN "Message is not valid."
        END IF
    ELSE
        RETURN "User not authorized to send message."
    END IF
  ELSE
    RETURN "Invalid sender or receiver."
  END IF
END FUNCTION
```

4.Pseudocode for Exercise Plan Creation

```
FUNCTION createExercisePlan(student, physiotherapist, exercises):
  IF isValidUser(student) AND isValidUser(physiotherapist) THEN
    IF isUserAuthorized(physiotherapist) THEN
        IF isExercisesValid(exercises) THEN
            exercisePlan = generateExercisePlan(exercises)
            storeExercisePlan(student, physiotherapist, exercisePlan)
            RETURN SUCCESS
        ELSE
            RETURN "Invalid exercises."
        END IF
    ELSE
        RETURN "Physiotherapist not authorized to create exercise plan."
    END IF
  ELSE
    RETURN "Invalid student or physiotherapist."
```

END IF

END FUNCTION

4.5 Interface Design

Figure 19 Registration Interface

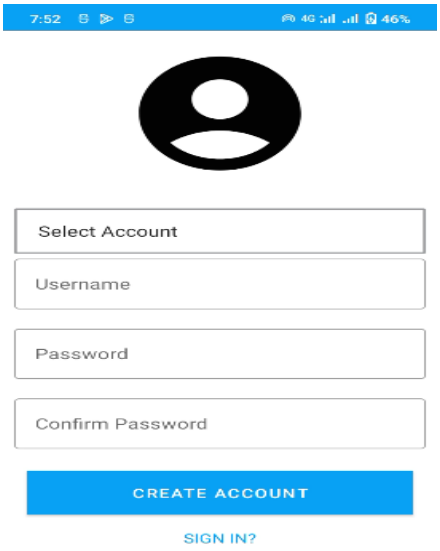
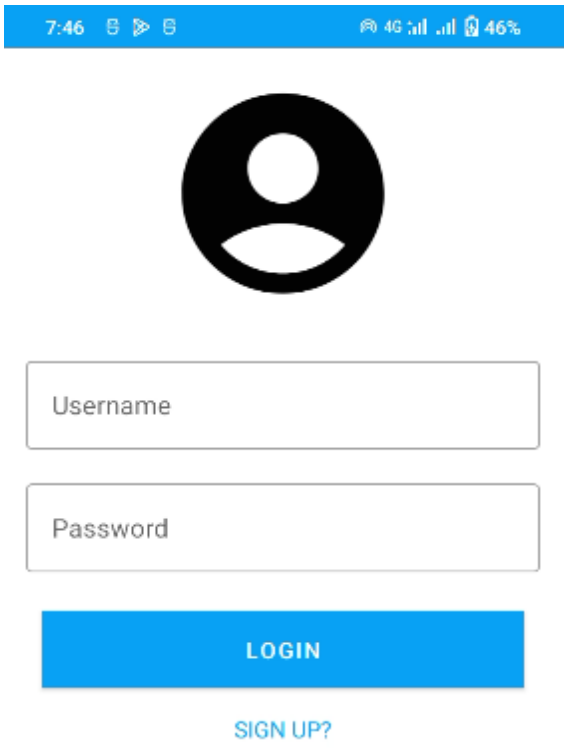


Figure 20 Sign Up Interface



The interface for registration and login is similar, User select which type of user are: either Physiotherapist or Student.

Figure 21 User Interface for Student

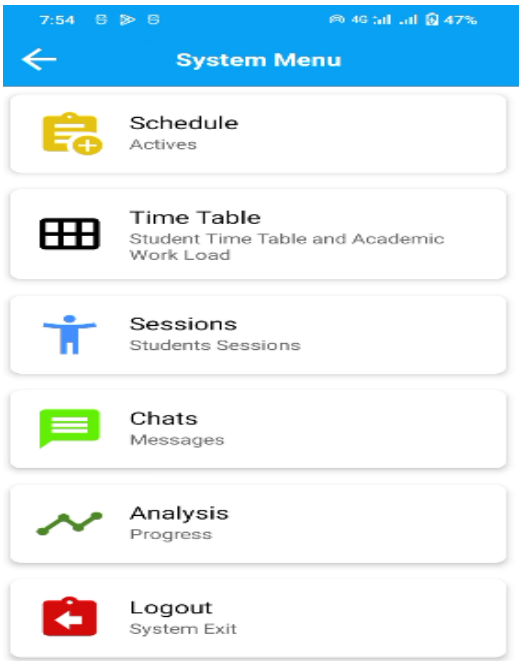
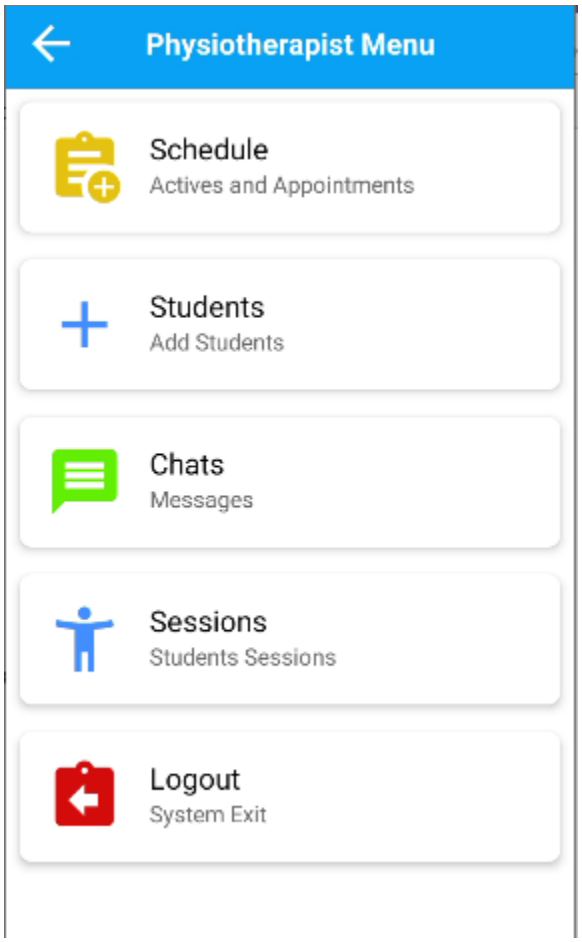
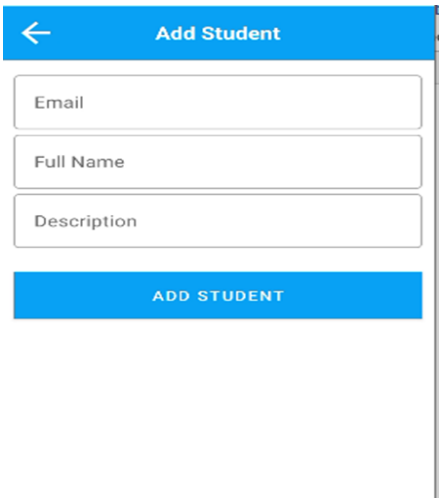


Figure 22 Physiotherapist User Interface



Physiotherapist can add their Students, through the Students tab

Figure 23 Physiotherapist Add Student



CHAPTER FIVE: IMPLEMENTATION & TESTING

5.1 Introduction

The implementation and testing phase of software development marks a crucial stage in the lifecycle of any system. This phase involves translating the design specifications into executable code, followed by rigorous testing to ensure that the system functions as intended and meets the defined requirements. In this section of the documentation, an overview of the implementation approach adopted for the proposed specialized scheduler and notification system, along with details of the testing methodologies employed to validate its functionality and performance.

5.2 Sample code for Scheduling Session/Appointment

```
package com.example.orcas;

import androidx.appcompat.app.AppCompatActivity;
import androidx.cardview.widget.CardView;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Toast;

public class menu extends AppCompatActivity {

    String user="";
    CardView schedule,sessionsview,chat,logoutnow, timetable, myana;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_menu);

        user=getIntent().getStringExtra("user");
        schedule=findViewById(R.id.schedule);
        schedule.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent1 = new Intent(menu.this, Schedule.class);
                intent1.putExtra("user", user);
                startActivity(intent1);
            }
        });

        timetable=findViewById(R.id.timetable);
```

```
timetable.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        Intent intent1 = new Intent(menu.this, TimeTable.class);
        intent1.putExtra("user", user);
        startActivity(intent1);
    }
});
sessionsview=findViewById(R.id.sessionsview);
sessionsview.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        Intent intent1 = new Intent(menu.this, StudentSessions.class);
        intent1.putExtra("user", user);
        startActivity(intent1);
    }
});

chat=findViewById(R.id.chat);
chat.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {

        Intent intent1 = new Intent(menu.this, StudentChat.class);
        intent1.putExtra("user", user);
        startActivity(intent1);
    }
});
logoutnow=findViewById(R.id.logoutnow);
logoutnow.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {

        Intent intent1 = new Intent(menu.this, MainActivity.class);
        startActivity(intent1);
    }
});

myana=findViewById(R.id.myana);
myana.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {

        Intent intent1 = new Intent(menu.this, analysis.class);
```

```
            intent1.putExtra("user", user);
            intent1.putExtra("id", "");
            intent1.putExtra("name", "");
            intent1.putExtra("reff", "");
            startActivity(intent1);
        }
    });
}
private void notifyUser(String message) {
    Toast.makeText(this, message, Toast.LENGTH_LONG).show();
}
public void back(View view) {

    Intent intent1 = new Intent(menu.this, Dashboard.class);
    intent1.putExtra("data", "1");
    intent1.putExtra("user", user);
    startActivity(intent1);
}
}
```

Code for Sending of Notifications

```
package com.example.orcas;

import android.Manifest;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.DialogInterface;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.graphics.Color;
import android.os.AsyncTask;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ListView;
import android.widget.SimpleAdapter;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
```

```
import androidx.annotation.NonNull;
import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.Properties;
import java.util.Timer;
import java.util.TimerTask;

import javax.mail.Authenticator;
import javax.mail.Message;
import javax.mail.MessagingException;
import javax.mail.PasswordAuthentication;
import javax.mail.Session;
import javax.mail.Transport;
import javax.mail.internet.InternetAddress;
import javax.mail.internet.MimeMessage;

public class DoctorDashboard extends AppCompatActivity {

    Timer timer;
    TimerTask timerTask;
    int time=0,one=0;
    String saved="", user="";

    ListView schedule;
    private static final int PERMISSION_REQUEST_CODE = 123;
    AlertDialog uploadingdialog,usersms;
    databasehelper dbHelper = new databasehelper(DoctorDashboard.this);
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
```

```
        setContentView(R.layout.activity_doctor_dashboard);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;


        });

        schedule=findViewById(R.id.schedule);
        fetchSchedules();


        String reNumber=getIntent().getStringExtra("data");
        one=Integer.parseInt(reNumber);
         user=getIntent().getStringExtra("user");

        if(time==0)
        {
            timer = new Timer();
            startTimer();

        }
        else {
            timerTask.cancel();
            timer = new Timer();
            startTimer();
        }

        if(Build.VERSION.SDK_INT>=Build.VERSION_CODES.TIRAMISU)
        {
            if(ContextCompat.checkSelfPermission(DoctorDashboard.this,
android.Manifest.permission.POST_NOTIFICATIONS)!= PackageManager.PERMISSION_GRANTED)
            {
                ActivityCompat.requestPermissions(DoctorDashboard.this, new
String[]{android.Manifest.permission.POST_NOTIFICATIONS},101);
            }
        }
        // makenotification();
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
            checkAndRequestPermission();
        }
```

```
}



SimpleAdapter ad;
private void  fetchSchedules()
{

    schedule=findViewById(R.id.schedule);
    List<Map<String,String>> MyDataList=null;
    MyDataList= dbHelper.getschedules();
    String[] Fromw={"heading","des","date","time", "textViewWithBorder"};
    int[] Tow={R.id.heading,R.id.des,R.id.date,R.id.time, R.id.textViewWithBorder};
    ad= new SimpleAdapter(DoctorDashboard.this,MyDataList,R.layout.schedules,Fromw,Tow);
    schedule.setAdapter(ad);
    schedule.setOnItemClickListener(new AdapterView.OnItemClickListener() {
        @Override
        public void onItemClick(AdapterView<?> adapterView, View view, int i, long l) {
            // notifyUser( dbHelper.scheduleid[i]+"");
            DisplayMassage("Do you want to Action this Schedule",dbHelper.scheduleid[i]+"");
        }
    });



}

private void notifyUser(String message)
{
    Toast.makeText(this, message, Toast.LENGTH_LONG).show();
}


private void  message(String sms, String ID) {

    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    View view = getLayoutInflater().inflate(R.layout.usermessage, null);
    builder.setView(view);
    TextView name=view.findViewById(R.id.usermessage);
    name.setText(sms);
    builder.setTitle("Message")

            .setNegativeButton("Delete", new DialogInterface.OnClickListener() {
```

```
                @Override
                public void onClick(DialogInterface dialogInterface, int i) {

                    dbHelper.drop("delete from schedule where ID='"+ID+"'");
                    fetchSchedules();


                }
            })
            .setPositiveButton("Edit", new DialogInterface.OnClickListener() {
                @Override
                public void onClick(DialogInterface dialogInterface, int i) {

                    Intent intent1 = new Intent(DoctorDashboard.this, DoctorUpdateSchedule.class);
                    intent1.putExtra("data", ID);
                    intent1.putExtra("user", user);
                    startActivity(intent1);

                }
            });

        usersms = builder.create();


    }


    @RequiresApi(api = Build.VERSION_CODES.S)
    private void checkAndRequestPermission() {
        if (ContextCompat.checkSelfPermission(this,
android.Manifest.permission.SCHEDULE_EXACT_ALARM)
                != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                new String[]{Manifest.permission.SCHEDULE_EXACT_ALARM},
                PERMISSION_REQUEST_CODE);
        }
    }

    // Handle the result of the permission request
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == PERMISSION_REQUEST_CODE) {
```

```
        if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
            // Permission granted, you can now set the alarm

        } else {
            // Permission denied, handle accordingly
        }
    }
}



private void startTimer()
{
    timerTask = new TimerTask() {
        @Override
        public void run() {
            runOnUiThread(new Runnable() {
                @Override
                public void run() {
                    time++;
                    Date currentTime = new Date();

                    SimpleDateFormat dateFormat = new SimpleDateFormat("hh:mm a",
Locale.getDefault());

                    if(dbHelper.startSchedule(getTodayDate(),dateFormat.format(currentTime)).equals(""))
                    {
                        one=0;
                    }
                    else
                    {
                        if(one==0)
                        {
                            one=1;
                            makenotification(dbHelper.heading, dbHelper.desc);
                            DoctorDashboard.sendemail doLogin = new DoctorDashboard.sendemail();
                            doLogin.execute("");


                        }
```

```
                }



            }
        });

    }
};
timer.scheduleAtFixedRate(timerTask,0,1000);
}

private String getTodayDate()
{
    return  new SimpleDateFormat("yyyy-MM-dd", Locale.getDefault()).format(new Date());
}


public void makenotification(String heading, String Message)
{
    String chanelID="CHANNEL_ID_NOTIFICATION";
    NotificationCompat.Builder builder=new
NotificationCompat.Builder(getApplicationContext(),chanelID);
    builder.setSmallIcon(R.drawable.notifications)
            .setContentTitle(heading)
            .setContentText(Message)
            .setAutoCancel(true)
            .setPriority(NotificationCompat.PRIORITY_DEFAULT);
    Intent intent= new Intent(getApplicationContext(), Dashboard.class);
    intent.addFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP);
    intent.putExtra("data", "1");

    PendingIntent pendingIntent= PendingIntent.getActivity(getApplicationContext(),
            0,intent, PendingIntent.FLAG_MUTABLE);
    builder.setContentIntent(pendingIntent);
    NotificationManager notificationManager=(NotificationManager)
getSystemService(Context.NOTIFICATION_SERVICE);

    NotificationChannel notificationChannel= null;
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        notificationChannel = notificationManager.getNotificationChannel(chanelID);
    }
    if(notificationChannel==null)
```

```
    {
        int importance =NotificationManager.IMPORTANCE_HIGH;
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            notificationChannel= new NotificationChannel(chanelID, "Some Description", importance);
        }
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            notificationChannel.setLightColor(Color.GREEN);
        }
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            notificationChannel.enableVibration(true);
        }
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            notificationManager.createNotificationChannel(notificationChannel);
        }
    }
    notificationManager.notify(0,builder.build());
}



public class sendemail extends AsyncTask<String, String, String>
{

    @Override
    protected void onPreExecute()
    {
        notifyUser("Sending email notification");


    }



    protected String doInBackground(String... strings) {

        String msg="";

        try
        {


            final String username = "taurayinhiwatiwa47@gmail.com";
            final String password = "xtzfxyqtotkufflf";
```

```
        Properties props = System.getProperties();
        props.put("mail.smtp.host", "smtp.gmail.com");
        props.put("mail.smtp.port", "465");
        props.put("mail.smtp.ssl.enable", "true");
        props.put("mail.smtp.auth", "true");




        javax.mail.Session session = Session.getInstance(props,
                new Authenticator() {
                    protected PasswordAuthentication getPasswordAuthentication() {
                        return new PasswordAuthentication(username, password);
                    }
                });


        MimeMessage message = new MimeMessage(session);
        message.addRecipients(Message.RecipientType.TO,
InternetAddress.parse(dbHelper.selectemail()));
        message.setSubject(dbHelper.heading);
        message.setText(dbHelper.desc);

        Thread thread= new Thread(new Runnable() {
            @Override
            public void run() {
                try {
                    Transport.send(message);

                } catch (MessagingException e) {
                    throw new RuntimeException(e);
                }
            }
        });



        thread.start();

        msg="Email sent successfully";
```

```
        }
        catch (Exception ex)
        {
            msg="Error: Check Your Connection";
        }




        return msg;
    }
    protected void onPostExecute(String r)
    {
        notifyUser(r);
    }


}


private  void DisplayMassage(String sms, String ID)
{


    message(sms, ID);
    usersms.show();




}

public void menu(View view) {
    Intent intent1 = new Intent(DoctorDashboard.this, DoctorMenu.class);
    intent1.putExtra("user", user);
    startActivity(intent1);


}
}
```

5.3 Software Testing

5.3.1 Unit Testing

Unit Testing Unit testing is a software testing technique where individual units or components of a software application are tested in isolation from the rest of the system. In this Specialized Scheduler and Notification mobile application individual components are tested to evaluate their performance. Each individual component, the scheduling of a session after considering the students timetable, the interface for both the physiotherapist and the student was tested, sending of scheduled notifications was tested. Each unit was performing well no bugs were found in each unit.

5.3.2 Module Testing

Module testing ensures the quality, reliability, and functionality of individual units or components within a software system. It involves testing each module in isolation from the rest of the system to validate its behavior according to the specified requirements and design. Module testing ensures that each component of the Specialized Scheduler and Notification System, such as appointment scheduling, communication channels, reminder systems, and exercise planning, functions correctly and meets the specified requirements. It also verifies the user-friendliness of the system's interface, regarding scheduling appointments based on academic schedules, facilitating communication between students and physiotherapists, and accessing personalized exercise plans.

Testing Objectives

<u>Usability Testing</u>

- Objective: To ensure that the Scheduler and Notification System provides a user-friendly interface for scheduling appointments, communicating with physiotherapists, and accessing exercise plans.

<u>Functionality Testing</u>

- Objective: To verify that all functionalities of the Scheduler and Notification System work as intended, meeting the specified requirements and user expectations.

<u>Communication Testing</u>

- Objective: To confirm that the communication channels within the Scheduler and Notification System facilitate seamless and effective communication between students and physiotherapists.

Reminder System Testing

- Objective: To ensure that the reminder system effectively notifies students and physiotherapists of upcoming appointments and activities.

Exercise Planning Testing

- Objective: To validate the creation, sharing, and access of personalized exercise plans tailored to each student's needs.

Test Cases:

Test Case 1: Usability Testing

Test ID: USABILITY-001

Description: Verify the usability of the appointment scheduling interface for Physiotherapist.

Steps:

- Log in to the Scheduler and Notification System as a physiotherapist.
- Navigate to the appointment scheduling section.
- Attempt to schedule a new appointment with a student.
- Expected Outcome: The scheduling interface should be intuitive and user-friendly, allowing physiotherapist to easily select appointment slots after viewing student's academic schedule.
- Actual Outcome: The scheduling interface is straightforward and allows physiotherapist to schedule appointments seamlessly.

Test Case 2: Functionality Testing

Test ID: FUNCTIONALITY-001

Description: Verify the functionality of automated reminders for upcoming appointments.

Steps:

- Physiotherapist schedule an appointment with the student
- Wait for the reminder time to elapse.
- Expected Outcome: An automated reminder email or push notification should be sent to both the student and physiotherapist, containing relevant details of the scheduled appointment.
- Actual Outcome: Automated reminders are successfully sent to both parties, reminding them of the upcoming appointment.

Test Case 3: Communication Testing

Test ID: COMMUNICATION-001

Description: Validate the messaging functionality between students and physiotherapists.

Steps:

- Log in as a physiotherapist and initiate a message to a student.
- Log in as the student and respond to the message.
- Expected Outcome: Messages should be delivered promptly between users, and both parties should be able to engage in a conversation seamlessly.
- Actual Outcome: Messages are delivered in real-time, and users can communicate effectively within the system.

Test Case 4: Exercise Planning Testing

Test ID: EXERCISE-001

Description: Ensure the creation and sharing of personalized exercise plans.

Steps:

- Physiotherapist creates a customized exercise plan for a student.
- Student accesses the exercise plan shared by the physiotherapist.
- Expected Outcome: The physiotherapist should be able to create tailored exercise plans, and students should securely access and view their personalized plans within the system.

- Actual Outcome: Physiotherapists can create exercise plans, and students can easily access and view their personalized plans.

Test Case 5: Progress Tracking Testing

Test ID: PROGRESS-001

Description: Validate the recording and visualization of progress of physiotherapy sessions attendance.

Steps:

- Physiotherapy sessions are conducted over a period, and progress of attendance is recorded.
- Users access the progress tracking feature to view their recorded data.
- Expected Outcome: Progress data, including sessions attended and those not attended, should be accurately recorded and visualized for users to track their physiotherapy progress effectively.
- Actual Outcome: Progress data is accurately recorded and visualized, enabling users to monitor their physiotherapy progress.

5.3.3 Integration testing

Integration testing involves testing the integration points between various modules or components of the Scheduler and Notification System to ensure they function correctly together as a unified system

Integration testing aims to achieve the following

1. Validate the interactions and data flow between modules, including appointment scheduling, communication channels, reminder systems, exercise planning, and progress tracking.
2. Ensure that interfaces between modules are implemented according to specifications and requirements.

The integration testing approach in the Specialized Scheduler and Notification is the Bottom-Up Integration Testing, which Starts with testing lower-level modules or components, gradually integrating higher-level modules.

Test Cases

Appointment Scheduling Module

Positive Test Case - Physiotherapist Scheduling:

- Input: Physiotherapist schedules a session with a student.
- Expected Output: Session is successfully scheduled and reflected in both the physiotherapist's and student's dashboard.

Negative Test Case - Conflict Resolution:

- Input: Physiotherapist schedules a session with a student during a time slot that overlaps with another existing appointment.
- Expected Output: System identifies the conflict and notifies the physiotherapist to reschedule the session.

Communication Module

Positive Test Case - Communication Integration:

- Input: Student sends a message to their physiotherapist for clarification.
- Expected Output: Physiotherapist receives the message and can respond accordingly within the system.

Negative Test Case - Message Delivery Failure:

- Input: Student sends a message to their physiotherapist, but there's an issue with the messaging system.
- Expected Output: System logs the issue and alerts the administrator while ensuring the message is not lost.

3. Reminder Notification Module:

Positive Test Case - Automated Reminder (Email):

- Input: Physiotherapy session is scheduled.
- Expected Output: Automated email reminder is sent to both the student and the physiotherapist prior to the session.

Positive Test Case - Automated Reminder (Push Notification):

- Input: Physiotherapy session is scheduled.
- Expected Output: Push notification reminder is sent to both the student and the physiotherapist prior to the session.

6. Progress Tracking Module

Positive Test Case - Progress Monitoring:

- Input: Physiotherapy session is completed.
- Expected Output: Both the student and the physiotherapist can track and monitor the student's attendance progress over time within the system.

The expected results for integration testing are the seamless functioning of all the integrated modules as a single system.

5.3.4 System testing

System testing is a level of software testing where the entire integrated system is tested as a whole. The purpose of system testing is to validate that the Specialized Scheduler and Notification system meets the specified requirements, functions as expected, and satisfies the needs of its users. The system testing is conducted in a controlled testing environment that replicates the production environment as closely as possible.

Test Plan

Functional Testing:

Objective: Verify that all functional requirements are implemented correctly and meet user expectations.

Test Cases

- Test appointment scheduling functionality, including conflict resolution.
- Test communication features such as messaging between students and physiotherapists.
- Test automated reminder notifications via email and push notifications.
- Test exercise plan creation, sharing, and accessibility.
- Test progress tracking and monitoring features.

Performance Testing:

Objective: Evaluate the performance of the system under normal and peak load conditions.

Test Cases:

- Test system response time for scheduling appointments and sending notifications.
- Test system scalability to accommodate a large number of users and appointments.
- Test system stability under sustained load.

Usability Testing

Objective: Assess the user-friendliness and intuitiveness of the system interface.

Test Cases

- Conduct user walkthroughs to evaluate the ease of scheduling appointments and communicating with physiotherapists.
- Gather feedback from users regarding the clarity of notifications and the accessibility of exercise plans.

Compatibility Testing

Objective: Ensure that the system is compatible with various devices, browsers, and operating systems.

Test Cases

- Testing the system on different operating systems (e.g., Windows, macOS, Linux).
- Testing the system on different devices (e.g., desktops, laptops, tablets, smartphones).

Test Execution

Performance Testing

1.Test System Response Time:

- Execution: Measure the time taken to schedule appointments and send notifications under normal and peak load conditions.

- Result: Response times are within acceptable limits under both normal and peak load conditions.
- Pass/Fail: Pass

Test System Scalability:

- Execution: Increase the number of concurrent users and appointments.
- Monitor system performance and response times.
- Result: The system scales effectively to accommodate increased users and appointments without significant degradation in performance.
- Pass/Fail: Pass

Usability Testing

Conduct User Walkthroughs

- Execution: Ask physiotherapist to schedule appointments and communicate with students while observing their interactions.
- Result: Physiotherapist find the scheduling process intuitive and can easily communicate with student.
- Pass/Fail: Pass

Compatibility Testing:

Testing on Different Devices

- Execution: Access the system using different devices including desktops, laptops, tablets, and smartphones.
- Result: The system is compatible with different devices and renders appropriately across various screen sizes without any issues.
- Pass/Fail: Pass

Database Testing

Database testing is a type of software testing that focuses on verifying the integrity, reliability, performance, and functionality of a database system.

Data Integrity Testing

- Objective: Ensure that data stored in the database is accurate, consistent, and free from errors.
- Execution: Insert sample data (student's schedule and timetable) into the database.
    - Retrieve and compare the inserted data with the expected values.
    - Result: Data retrieved matches the expected values, indicating data integrity.
    - Pass/Fail: Pass

Data Validation Testing

Objective: Verify that the database enforces proper data validation rules to prevent invalid data entry.

- Execution: Attempt to insert invalid data (e.g., incorrect data types, out-of-range values) into the database.
- Verify that the database rejects invalid data entries.
- Result: Invalid data entries are rejected by the database, ensuring data validity.
- Pass/Fail: Pass

Performance Testing

Objective: Evaluate the performance of database operations under normal and peak load conditions.

- Execution: Measure the time taken for common database operations such as data retrieval, insertion, and updating.
- Increase the workload on the system and observe any degradation in database performance.
- Result: Database operations are completed within acceptable time limits under both normal and peak load conditions.
- Pass/Fail: Pass

5.3.5 Acceptance testing

Acceptance testing, the system is evaluated to ensure that it meets the specified requirements and is ready for deployment. This testing phase involves validating the system's functionality, usability, performance, and other quality attributes from the perspective of end-users or stakeholders.

Acceptance testing for the Specialized Scheduler and Notification system ensures that it meets the specified requirements and functions as expected before it is deployed for use.

Acceptance testing plays a pivotal role in ensuring the quality and readiness of the system for deployment. By subjecting the system to rigorous testing before release, potential issues, defects, and discrepancies can be identified and addressed, thereby minimizing the risk of post-deployment failures and ensuring a smoother rollout.

Acceptance tests are designed to validate each requirement and ensure that the system behaves as expected, addressing the needs and expectations of users. Acceptance testing assesses the functionality of the system to ensure that it performs its intended tasks accurately and efficiently. This includes testing individual features, workflows, and use cases to verify that they work as intended and meet the needs of end-users.

User Interface Testing

- Verify that the user interface is intuitive, user-friendly, and accessible for physically challenged students and physiotherapists.
- Test navigation flows, input fields, buttons, and other UI elements to ensure they function correctly.
- Validate that the interface accommodates accessibility features such as screen readers and keyboard navigation.

Appointment Scheduling Testing

- Test the appointment scheduling functionality to ensure that physiotherapist can schedule sessions without conflicts with student academic schedules.
- Test the system's ability to send automated reminders for upcoming appointments.
- Verify that messages are delivered promptly and reliably.
- Verify that students can access their personalized exercise plans and view instructions correctly.

Accessibility Testing

- Test accessibility features to ensure that the system is usable for physically challenged students.

Performance Testing

- Conduct performance tests to ensure that the system can handle expected user loads without significant slowdowns or errors.
- Test response times for key actions such as appointment scheduling, message delivery, and data retrieval.

User Acceptance Testing (UAT)

- Conduct UAT with representative users, including physically challenged students and physiotherapists, to gather feedback and ensure that the system meets their needs and expectations.
- Incorporate user feedback and make necessary adjustments before final deployment.

CHAPTER SIX: CONCLUSIONS AND RECOMMENDATIONS

6.0 Introduction

In this chapter, the results and summary of the project is provided, including the recommendations and future of the system.

6.1 Results

Figure 24 Physiotherapy Scheduled Session by the Physiotherapist

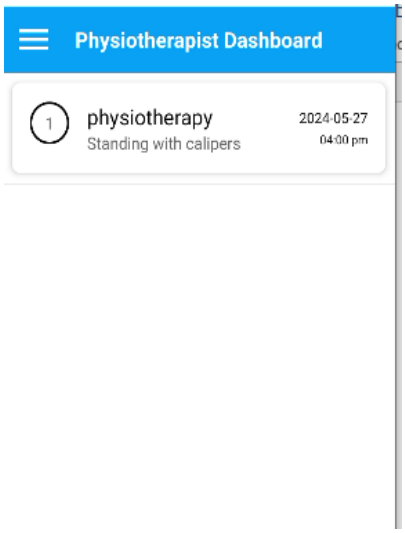


Figure 25 Scheduled Session on Student Dashboard

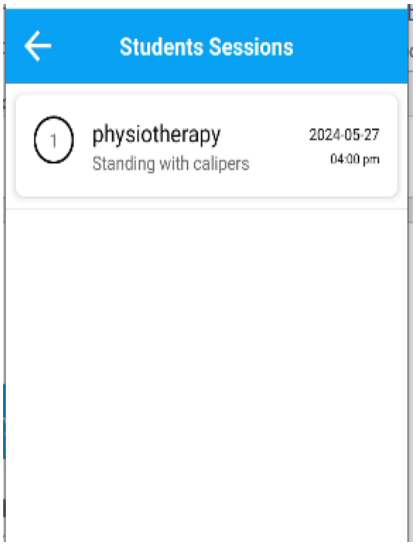
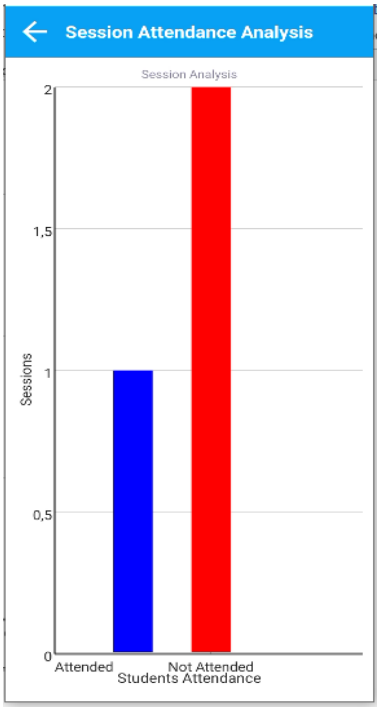


Figure 26 Visual Representation of progress



1. The specialized scheduler and notification system successfully implemented a user-friendly interface for appointment scheduling.
2. Physiotherapist Scheduling: Physiotherapists were able to easily schedule sessions with their patients/students using the system.
3. Communication Channel: The system facilitated seamless communication between students and physiotherapists. Students could easily reach out to their physiotherapists for clarifications, questions, or updates through the platform. This streamlined communication enhanced the overall experience for both parties, fostering a collaborative and supportive environment.
4. Automated Reminders: Automated reminders were implemented through email or push notifications to ensure students and physiotherapists were reminded of their upcoming physiotherapy sessions. This feature helped reduce the likelihood of missed appointments and improved overall adherence to physiotherapy session attendance.
5. Progress Tracking and Monitoring: Users could track and monitor their physiotherapy attendance progress through the system.

6.1.1 Summary

In conclusion, the specialized scheduler and notification system effectively addressed the objectives outlined for optimizing physiotherapy services for students. By incorporating features such as a user-friendly interface, streamlined communication channels, automated reminders, tailored exercise plans, and progress tracking capabilities, the system enhanced the overall experience for both students and physiotherapists.

The system not only facilitated efficient appointment scheduling but also fostered greater engagement and collaboration between students and physiotherapists.

Moving forward, continuous feedback and iterative improvements will be essential to further refine the system and adapt it to evolving needs and preferences. Overall, the specialized scheduler and notification system represented a significant advancement in optimizing physiotherapy services/ adherence within academic settings, ultimately promoting better health outcomes and improved quality of life for students.

6.2 Recommendations

1. Integrate the specialized scheduler and notification application with academic systems or student portals to automatically import class schedules, assignment deadlines, and exam dates. This ensures that physiotherapy sessions are scheduled around academic commitments
2. Provide physically challenged students with flexible rescheduling options in case of unexpected changes in their academic schedule or health condition. Allow them to easily reschedule physiotherapy sessions without penalty.
3. Provide alternative communication methods such as voice-to-text or text-to-speech for students with mobility or speech impairments.

6.3 Future Works

1. Integration with Wearable Devices – integrating the app with wearable devices such as fitness trackers or smartwatches to collect real-time data on students' physical activities and progress. This data can be used to further personalize exercise plans and track progress accurately.

2. Incorporate telehealth capabilities into the system, allowing for virtual physiotherapy sessions. This would enable remote consultations and exercises, expanding access to physiotherapy services for students who are unable to attend in-person sessions
3. Enhance the communication module with additional features such as video conferencing, instant messaging, or chatbot support
4. Implementing data analytics capabilities to analyze the collected data and generate insights

6.4 Conclusion

In conclusion, the proposed Specialized Scheduler and notification application offers a comprehensive solution for managing physiotherapy sessions, communication, and exercise plans for physically challenged students. By considering the students' academic schedule and workload system minimizes conflicts and optimizes session scheduling. The system's user-friendly interface, automated reminders, and secure communication channels enhance the overall experience for both physiotherapists and students. The personalized exercise plans and progress tracking features enable tailored care and informed treatment adjustments. The proposed system architecture ensures scalability, security, and maintainability, making it a viable solution for educational institutions. By streamlining physiotherapy management, this system has the potential to improve the lives of physically challenged students at King George VI center, enabling them to manage their conditions more effectively and achieve their academic goals. Future development directions include integrating with existing student information systems, expanding to cater to other healthcare professionals, and incorporating artificial intelligence to enhance exercise plan generation and progress

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Appendices A - User Manual

Introduction

Welcome to the Specialized Scheduler and Notification System manual. This system is designed to streamline the process of scheduling physiotherapy sessions for students while considering their academic schedules and ensuring effective communication between students and physiotherapists. The system also facilitates exercise plan creation, sharing, and progress tracking to enhance the physiotherapy experience.

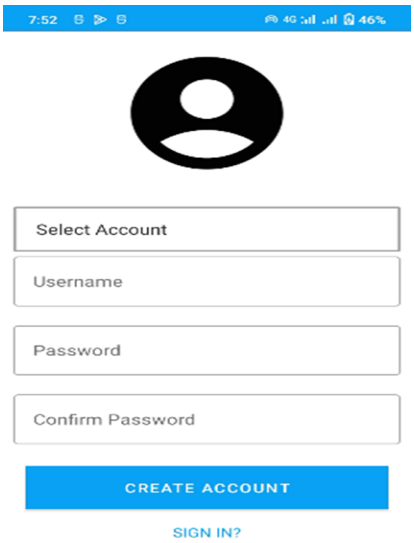
1. **System Overview**

The system consists of a mobile application available on android devices accessible to both students and physiotherapists. It offers features such as appointment scheduling, communication tools, automated reminders, exercise plan creation, and progress tracking.

2. **To Get Started with Specialized Scheduler and Notification App**

To get started with the Specialized Scheduler and Notification, users need to install the application file on their devices. Once installed, users need to create an account by providing their username (use valid email address as username) and password. After creating the account, users can log in the mobile application and start scheduling.

Figure 27 Sign In



3. **User Roles and Permissions**

Student: Can schedule academic workload, communicate with physiotherapists, access exercise plans, and track progress.

Physiotherapist: Can schedule sessions with students, communicate with them, create exercise plans, and monitor progress.

4. Interface Design

The user interface is designed to be intuitive and user-friendly, with easy navigation and clear labeling of features. It includes sections for scheduling appointments, communication, exercise plans, and progress tracking.

Figure 28 Student Interface

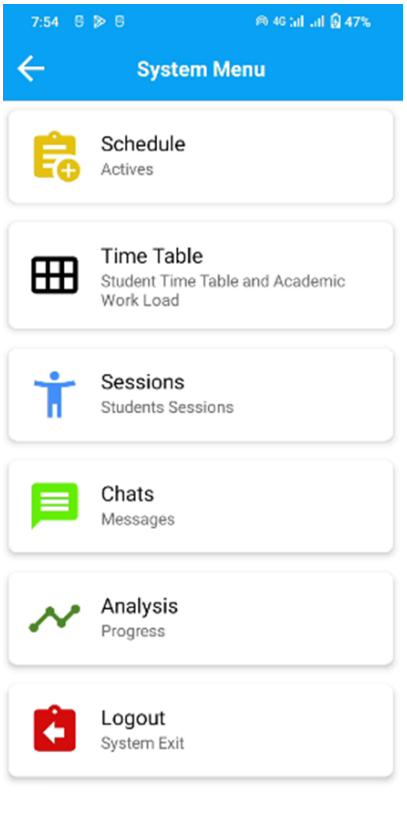
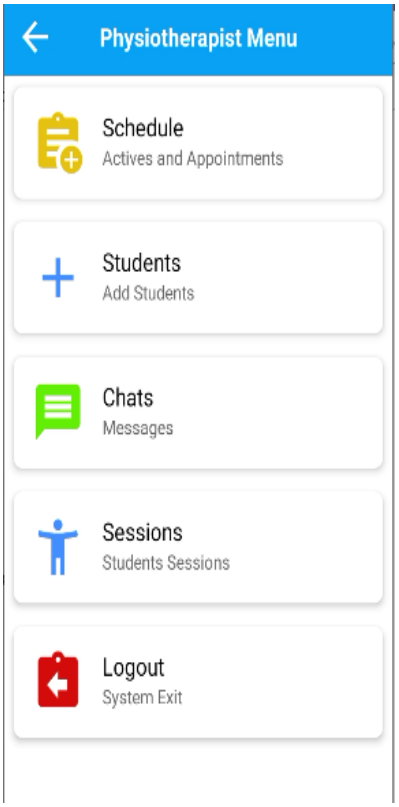
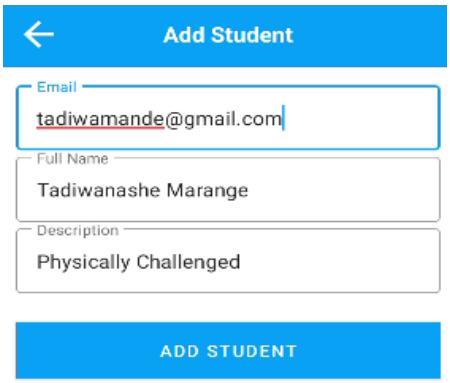


Figure 29 Physiotherapist Interface

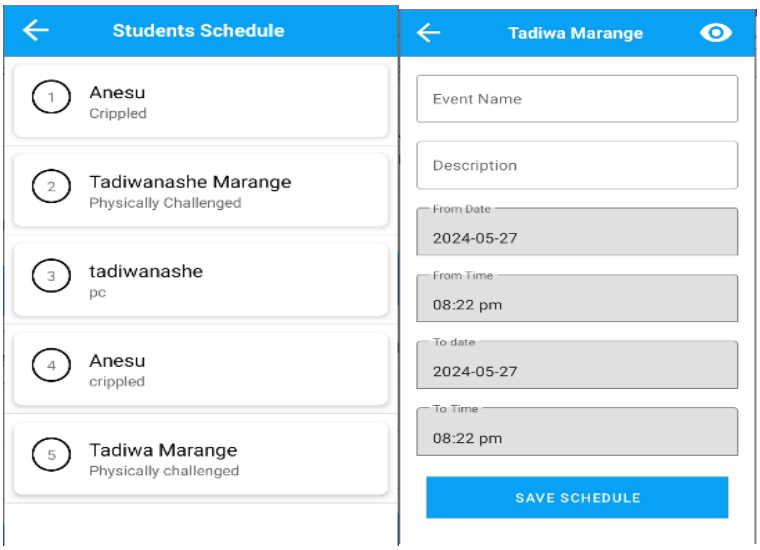


5. Scheduling Appointments

- Physiotherapist has to Add student First, with which they want to schedule sessions with

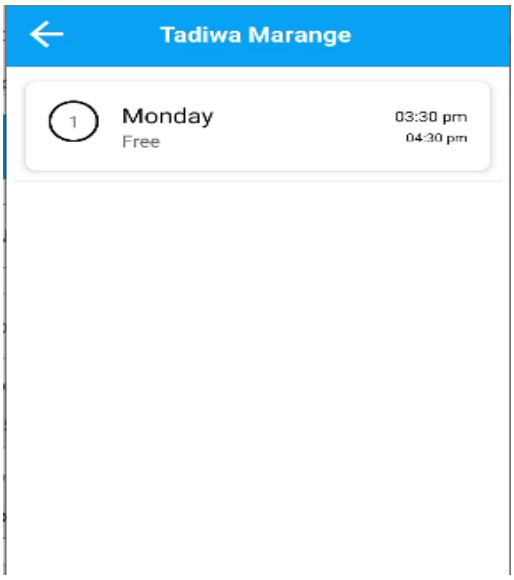


- Click the Schedule tab and Select the student the want to schedule with



Physiotherapist can view available appointment slots based on the student academic schedule and schedule sessions accordingly ensuring no conflicts with academic commitments.

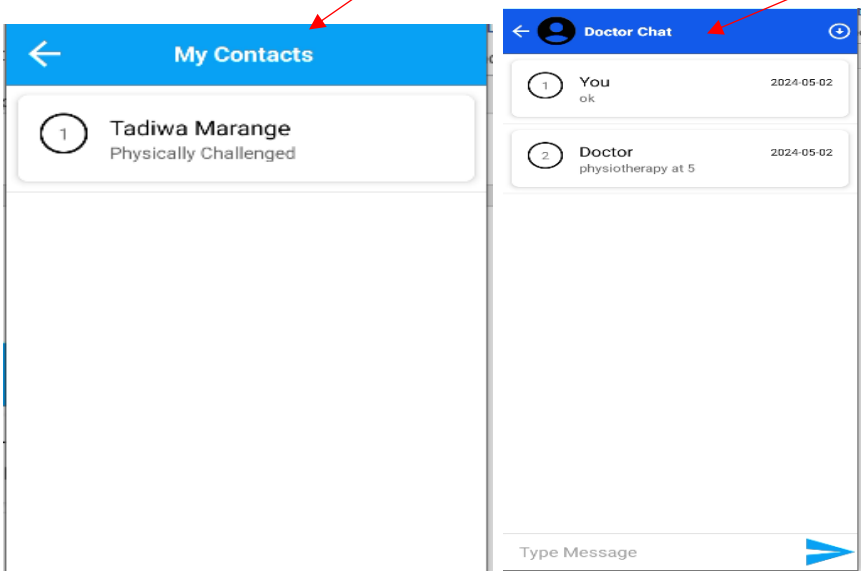
- Click On the (eye icon) to view student academic workload



6. Communication Features

The system provides messaging functionality for students and physiotherapists to communicate regarding appointments, clarifications, questions, or updates. This promotes seamless communication and ensures effective collaboration.

- Select the Chat Tab 1. (Physiotherapist Select the student they want to communicate with and initiate a message) 2. Student can also chat with the Physiotherapist



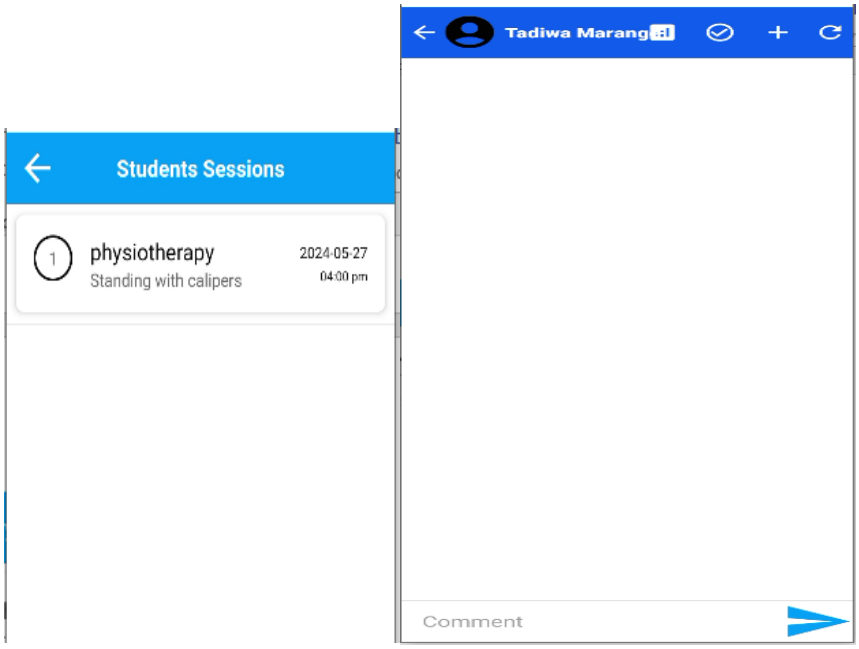
7. Automated Reminders

Email and push notifications are sent to students and physiotherapists to remind them of upcoming physiotherapy sessions. This helps in minimizing missed appointments and ensures both parties are well-prepared.

8. Exercise Plan Creation and Sharing:

Physiotherapists can create personalized exercise plans tailored to each student's needs. These plans can be easily shared with students through the platform, ensuring clear instructions for home exercises.

- Click the Session tab, Select the scheduled session and click the (+) to add the exercise plan for that session

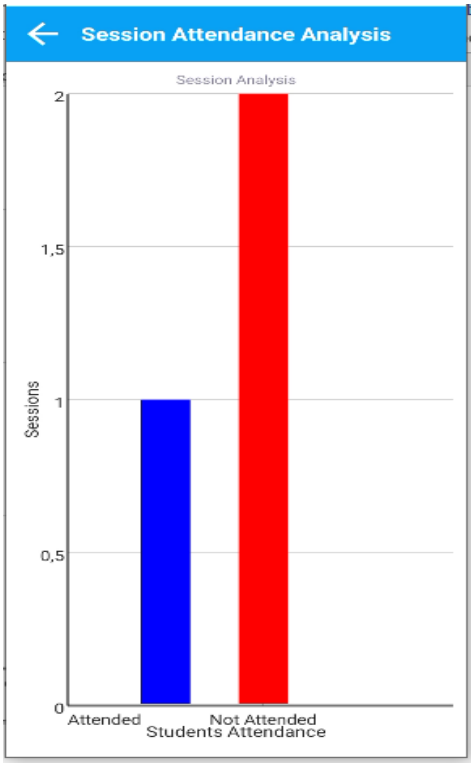


10. Progress Tracking and Monitoring

Both students and physiotherapists can track and monitor physiotherapy progress of attendance through the system.

- Student select Analysis Tab and view their progress (the attendance is marked by the physiotherapist after session attendance (Physiotherapist click the attend (the tick sign) button)

Student Progress (Interface) Physiotherapist click the attendance button



12. Conclusion

The Specialized Scheduler and Notification System is a comprehensive platform designed to enhance the physiotherapy experience for students and physiotherapists alike. With its user-friendly interface, effective communication tools, and advanced features, it aims to streamline the scheduling process, improve collaboration.

Appendices B – Technical Paper

Specialized Scheduler and Notification System

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***Abstract*- Balancing between physiotherapy sessions and school academic load can be challenging or difficult for students living with disabilities especially at King George VI center. To address this issue, we propose a Specialized Scheduler and Notification Mobile Application that allows students and physiotherapist to schedule therapy sessions without conflicting with the academic schedule. The system allows for students to enter their academic schedule, for physiotherapist to view student's free periods and schedule physiotherapy sessions at their free periods, automated reminders of sessions to be send to students and physiotherapist and for students to track their progress of session attendance. The application was built in android studio using Java. The proposed system will provide personalized notifications, scheduling capabilities, and collaboration features. It also supports time management, ensures accessibility, improves organization and planning, enables progress tracking and motivation, and enhances communication and support.**

Keywords: Scheduling, notifications, sessions, progress tracking, communication

I. INTRODUCTION

Students living with physical disabilities at King George VI center often face challenges in managing their healthcare needs (physiotherapy sessions) alongside their academic responsibilities. Physiotherapy sessions are crucial for their well-being, but scheduling and managing these sessions can be difficult due to conflicting academic schedules, forgetfulness, and a lack of personalized time management tools. The proposed tool is a Specialized Scheduler and Notification Mobile Application that allows students and physiotherapist to schedule therapy sessions without conflicting with the academic schedule

II. PROBLEM STATEMENT

Physically challenged students living with disabilities at King Georges VI center often face challenges in managing their time and balancing their school work with physiotherapy sessions. The tend to focus more on their school work load neglecting exercises especially students living with paralysis. They require a system that can provide timely notifications and assist in scheduling their activities effectively. The existing solutions lack customization and fail to cater to the specific needs of these students. Therefore, there is a need for a scheduler and notification application that is specifically designed for these students living with physical disabilities.

III. RELATED WORKS

The SnoopMe Interactive Task Scheduler mobile application[1] by A. Dalvi and I. Siddavatam was designed to assist students in managing their tasks effectively by providing alerts and monitoring their mobile usage. It allows students to schedule tasks, set reminders, and receive notifications about task completion. The app tracks mobile activities and recommends usage restrictions based on the data collected. Students can benefit from graphical representations of their time spent on different apps. Feedback from users indicates positive experiences with

the app, with suggestions for additional features to enhance its effectiveness as a student-centric scheduler.

The InvigilatorApp[8], a system designed for efficient scheduling and reminder notifications for invigilation duties in educational institutions. The system automates and optimize the scheduling process, improve communication between administrators and invigilators, and streamline exam administration. Key features include web-based admin and faculty dashboards, real-time notifications, round-robin scheduling, and database management. The system utilizes algorithms such as Particle Swarm Optimization, Constraint Programming, Ant Colony Optimization, Tabu Search, and Hybrid Approaches to enhance scheduling efficiency. The system architecture involves React JavaScript for the user interface and Django framework for the backend, ensuring timely reminders and email notifications.

In [4] an android application designed by D. Ameta, K. Mudaliar, and P. Patel, is focusing on enhancing interactions between doctors and patients while promoting adherence to medication schedules. The application includes a feature that automatically sets off alarms to remind users of their medication timings. Additionally, it allows patients to search for healthcare providers, access health-related information, and receive notifications via email or messages. The system aims to be cost-effective, efficient, and supportive of medication adherence by offering features like facilitating communication between doctors and patients, assisting in medication scheduling, and presenting a user-friendly interface that does not necessitate specialized hardware for operation.

IV. SOLUTION

To solve the problems of the current system of scheduling and notification, A specialized scheduler and notification mobile application is going to be developed with the following objectives:

1. To design a user-friendly interface for appointment scheduling that specifically considers the academic schedules of the student to avoid conflicts with physiotherapy sessions.
2. To allows the physiotherapists to schedule sessions with their patients/ students
3. To allows students to easily communicate with their physiotherapist for clarifications, questions, or updates
4. To allow automated reminders through email, or push notifications to ensure students and physiotherapist are reminded of their upcoming physiotherapy sessions.
5. To create exercise plans tailored to each student's needs which they can share with their patients
6. To allows physically challenged students to securely login and access their personalized physiotherapy exercise plans as shared by their physiotherapist
7. To allow users to track and monitor their physiotherapy progress to enable adjustments to their treatment plans as needed.

A. FEATURES OF THE SYSTEM

The proposed system will have the following features:

1. User-friendly interface for appointment scheduling that considers academic schedules to avoid conflicts with physiotherapy sessions.
2. Physiotherapists can schedule sessions with their patients/students.
3. Students can easily communicate with their physiotherapist for clarifications, questions, or updates.
4. Automated reminders through email or push notifications to ensure students and physiotherapists are reminded of their upcoming physiotherapy sessions.

5. Exercise plans tailored to each student's needs which they can share with their patients.
6. Physically challenged students can log in and access their personalized physiotherapy exercise plans as shared by their physiotherapist.
7. Users can track and monitor their physiotherapy progress to enable adjustments to their treatment plans as needed.

B. BENEFITS OF THE SYSTEM

The Proposed system will have the following benefits:

1. The system streamlines appointment scheduling
2. Improves communication between students and physiotherapists
3. Sends automated reminders
4. Allows personalized exercise plans
5. Users can track their physiotherapy progress, allowing physiotherapists to make necessary adjustments based on individual progress, resulting in more effective and personalized care.
6. The system enhances collaboration and attendance among students and physiotherapists.

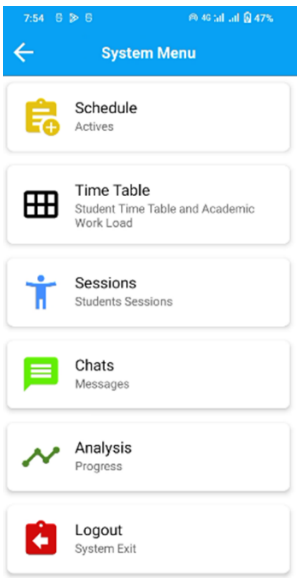
C. SOLUTION ARCHITECTURE

1. Presentation Layer- User Interface (UI) for appointment scheduling of the mobile application was developed using Java in android studio
2. Application Layer – Scheduling Module which manages appointment scheduling, conflicts, and availability, communication modules which facilitates secure messaging between physiotherapists and students. Notification module which sends automated reminders and notifications
3. Business Logic Layer – Triggers reminders and notifications based on schedules and deadlines
4. Data Access Layer – Database (MSSQL) which stores user data, schedules, exercise plans and communication records
5. Infrastructure Layer- Server (on-premises (SQL Server Management Studio Management 19)) hosts the database, notification services (email notifications) to send reminders and notification.
6. Security – Data Encryption (HTTPS) which protects data in transit and at rest

V. RESULTS AND FUTURE WORKS

A. RESULTS

Fig 21 Student User Interface



The system performed well in allowing the physiotherapist to see the student's academic load and schedule for physiotherapy sessions, sending automated reminder to both the student and the physiotherapist, also allowing for communication between the physiotherapist and the student through a messaging platform inbuilt in the platform and the results are summarized in Table I.

Objectives	Fully Achieved	Partially Achieved
To design a user-friendly interface for appointment scheduling that specifically considers the academic schedules of the student to avoid conflicts with physiotherapy sessions.	✓	
To allows the physiotherapists to schedule sessions with their patients/ students	✓	
To allows students to easily communicate with their physiotherapist for clarifications, questions, or updates	✓	
To allow automated reminders through email, or push notifications to ensure students and physiotherapist are reminded of their upcoming physiotherapy sessions.	✓	
To create exercise plans tailored to each student's needs which they can share with their patients	✓	
To allows physically challenged students to login and access their personalized physiotherapy exercise plans as shared by their physiotherapist	✓	
To allow users to track and monitor their physiotherapy progress to enable adjustments to their treatment		✓

plans as needed		

B. FUTURE WORKS

Future works of the system include:

1. Integration with Wearable Devices – integrating the app with wearable devices such as fitness trackers or smartwatches to collect real-time data on students' physical activities and progress. This data can be used to further personalize exercise plans and track progress accurately.
2. Incorporate telehealth capabilities into the system, allowing for virtual physiotherapy sessions. This would enable remote consultations and exercises, expanding access to physiotherapy services for students who are unable to attend in-person sessions
3. Enhance the communication module with additional features such as video conferencing, instant messaging, or chatbot support
4. Implementing data analytics capabilities to analyze the collected data and generate insights

VI. CONCLUSION

In conclusion, the proposed Specialized Scheduler and notification application offers a comprehensive solution for managing physiotherapy sessions, communication, and exercise plans for physically challenged students. By considering the students' academic schedule and workload system minimizes conflicts and optimizes session scheduling. The system's user-friendly interface, automated reminders, and secure communication channels enhance the overall experience for both physiotherapists and students. The personalized exercise plans and progress tracking features enable tailored care and informed treatment adjustments. The proposed system architecture ensures scalability, security, and maintainability, making it a viable solution for educational institutions. By streamlining physiotherapy management, this system has the potential to improve the lives of physically challenged students at King George VI center, enabling them to manage their conditions more effectively and achieve their academic goals. Future development directions include integrating with existing student information systems, expanding to cater to other healthcare professionals, and incorporating artificial intelligence to enhance exercise plan generation and progress tracking. The Scheduler and Notification Application demonstrates the potential of technology in enhancing healthcare management and promoting inclusive education.

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